

CSE 4621

Machine Learning

Topic: Convolutional Neural Networks II

Source: [DeepLearning.AI](https://deeplearning.ai)

Presented by

Rafsanjany Kushol

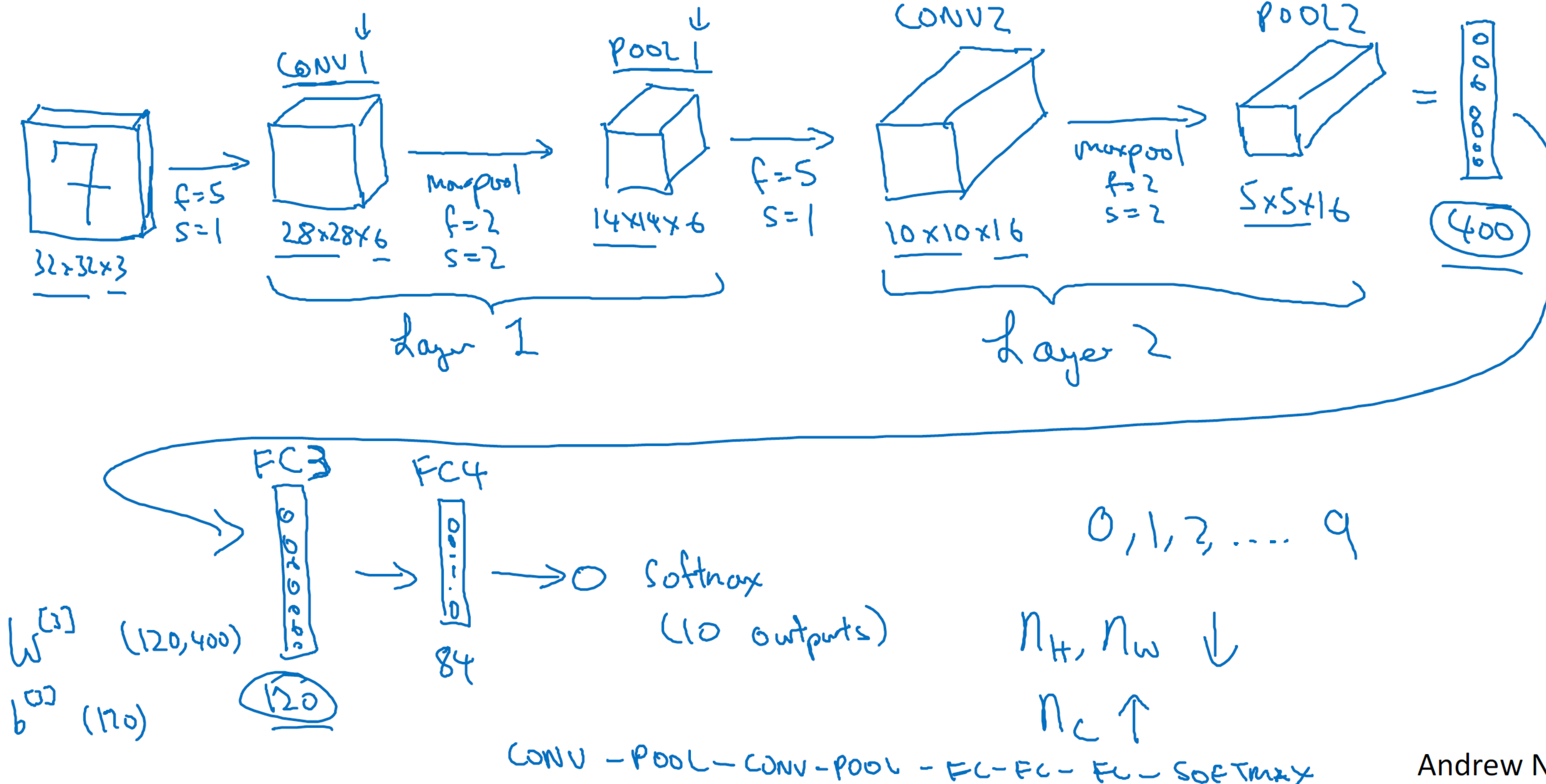
Assistant Professor



Types of layer in a convolutional network

- Convolution (CONV) ←
 - Pooling (POOL) ←
 - Fully connected (FC) ←
- 

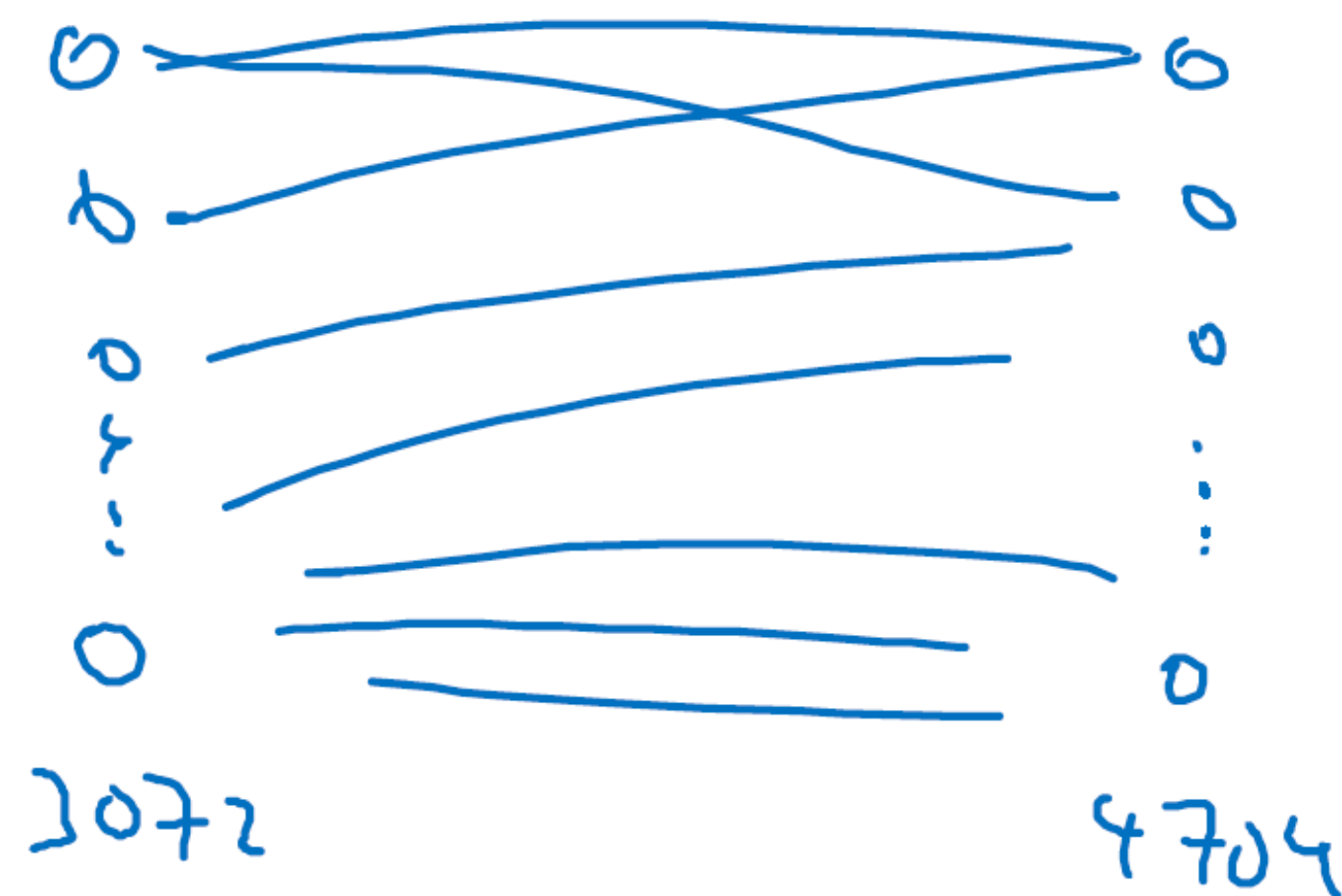
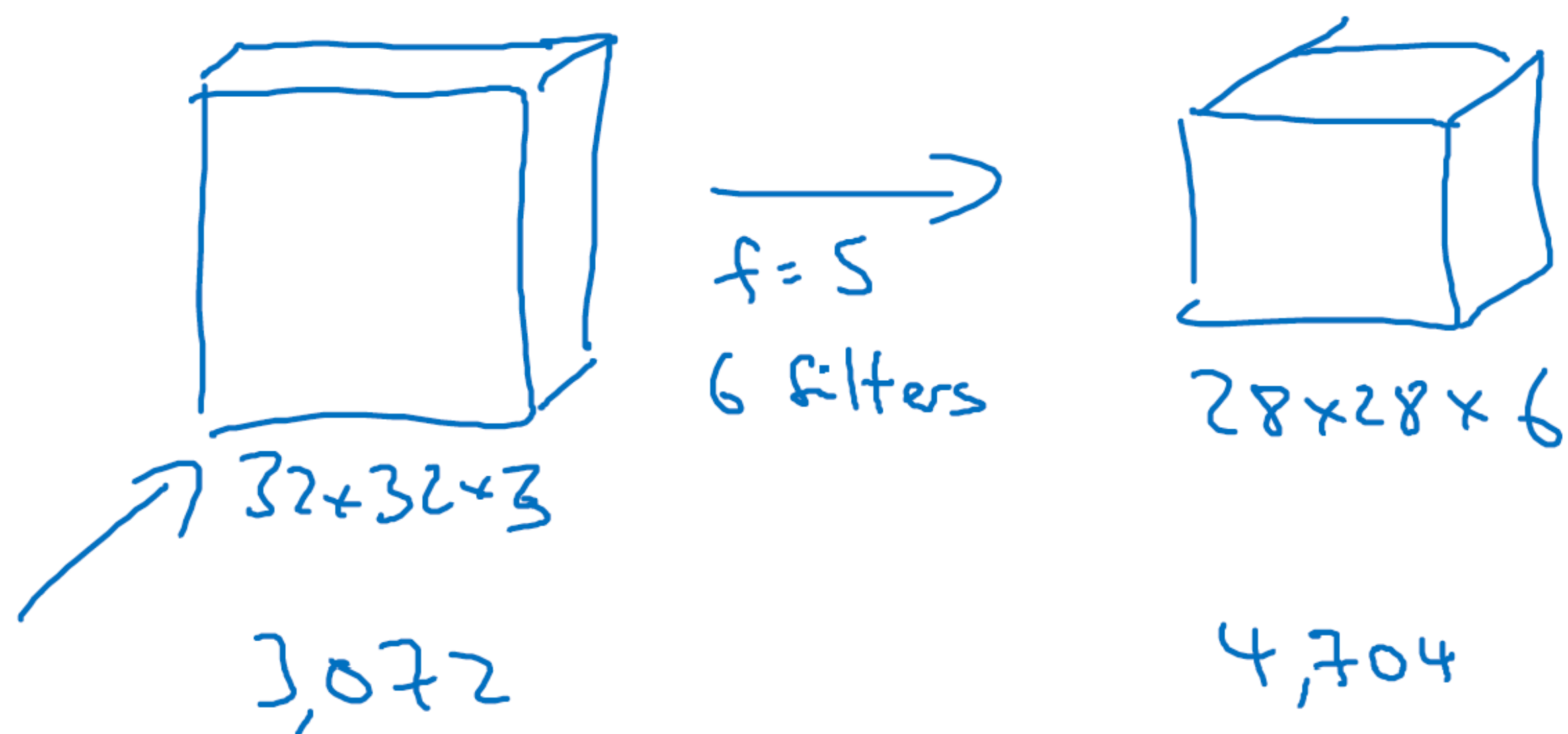
Neural network example



Neural network example

	Activation shape	Activation Size	# parameters
Input:	(32,32,3)	— 3,072 a^{col}	0
CONV1 (f=5, s=1)	(28,28,8)	<u>6,272</u>	208 ←
POOL1	(14,14,8)	<u>1,568</u>	0 ←
CONV2 (f=5, s=1)	(10,10,16)	<u>1,600</u>	416 ←
POOL2	(5,5,16)	<u>400</u>	0 ←
FC3	(120,1)	<u>120</u>	48,001 } ←
FC4	(84,1)	<u>84</u>	10,081 } ←
Softmax	(10,1)	<u>10</u>	841

Why convolutions



$$5 \times 5 = 25$$

$$26$$

$$6 \times 26 = 156 \text{ parameters}$$

$$3,072 \times 4,704 \approx \underline{14M}$$

Why convolutions

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0

*

1	0	-1
1	0	-1
1	0	-1

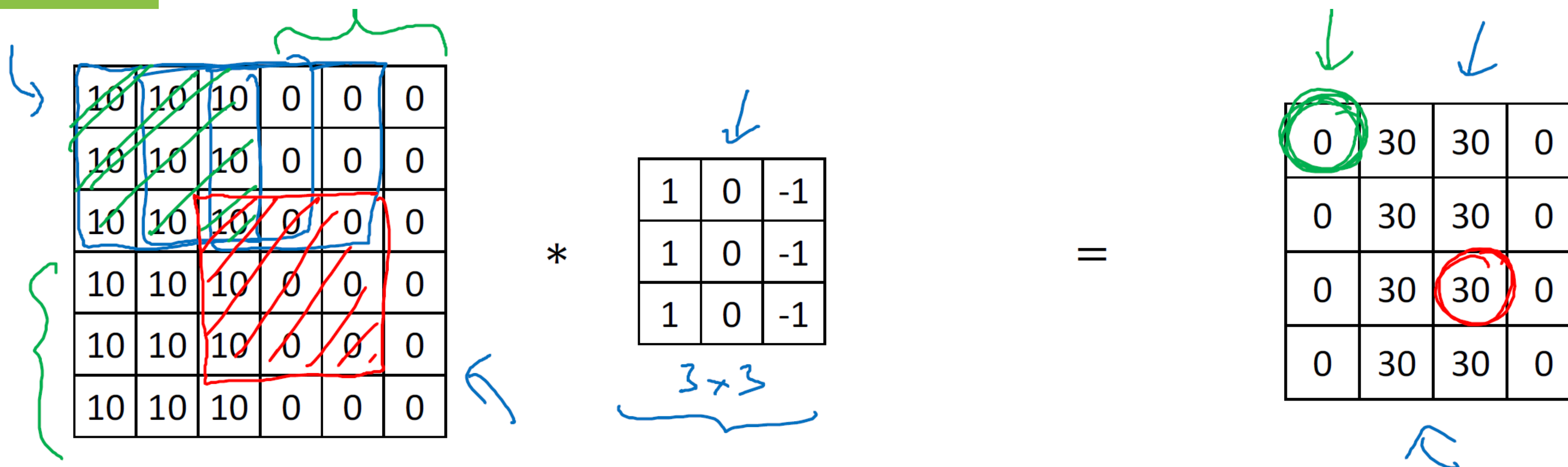
3x3

=

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0

Parameter sharing: A feature detector (such as a vertical edge detector) that's useful in one part of the image is probably useful in another part of the image.

Why convolutions

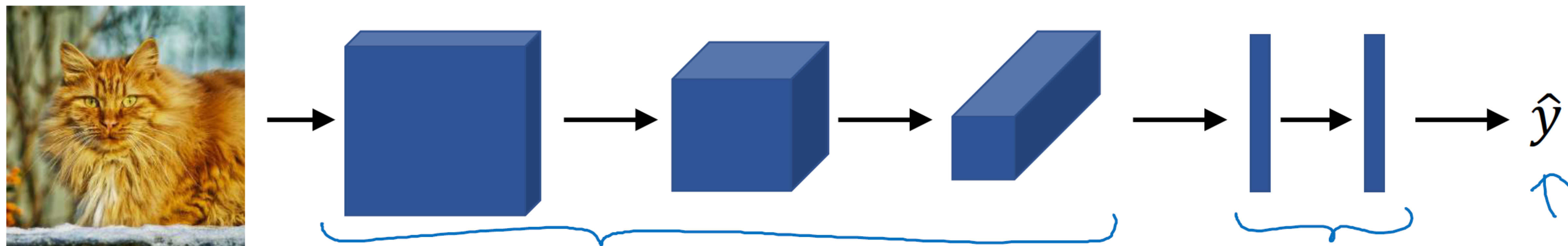


Parameter sharing: A feature detector (such as a vertical edge detector) that's useful in one part of the image is probably useful in another part of the image.

→ **Sparsity of connections:** In each layer, each output value depends only on a small number of inputs.

Putting it together

Training set $(x^{(1)}, y^{(1)}) \dots (x^{(m)}, y^{(m)})$.



$$\text{Cost } J = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$

Use gradient descent to optimize parameters to reduce J

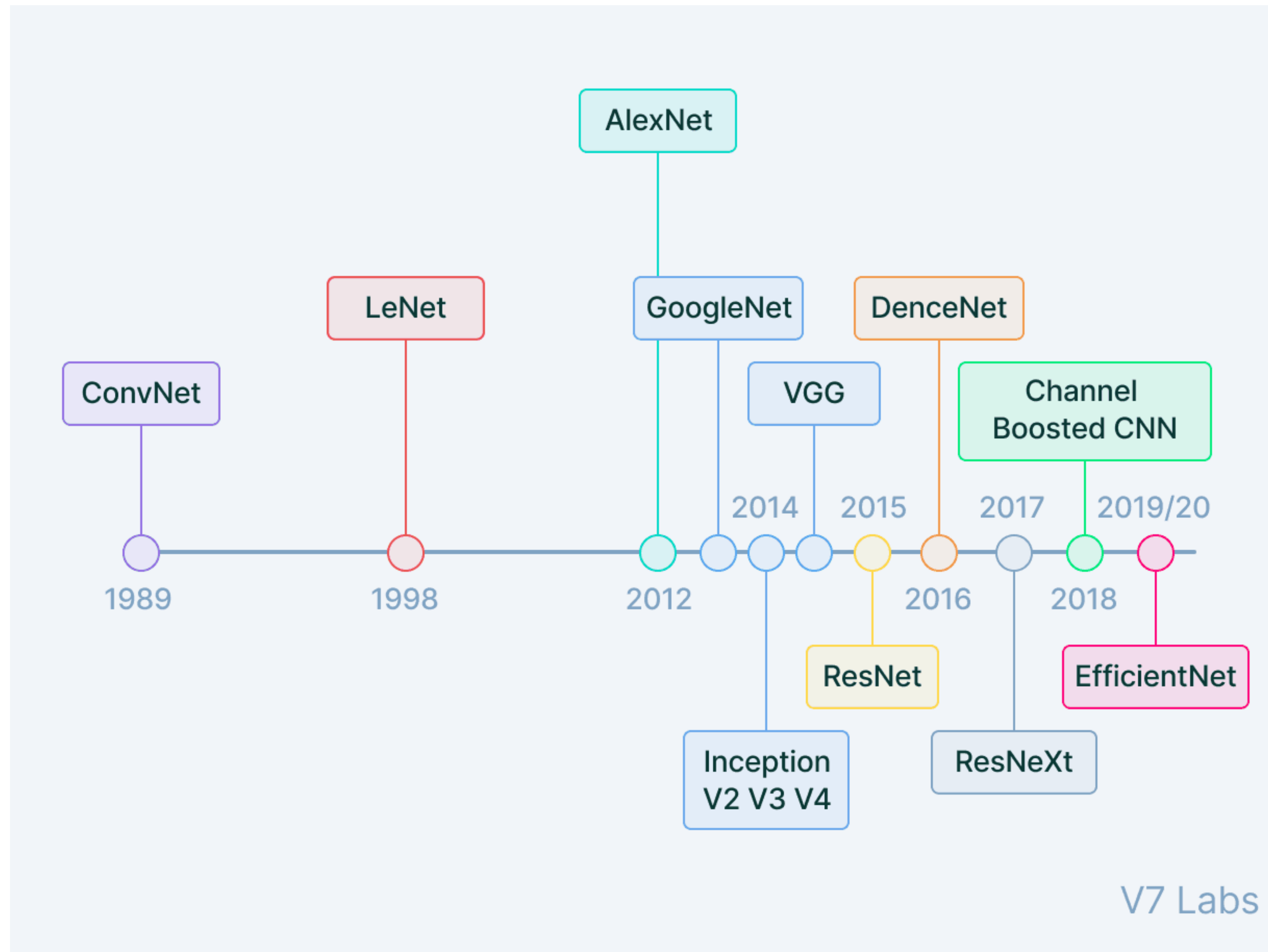
Case studies

Classic networks:

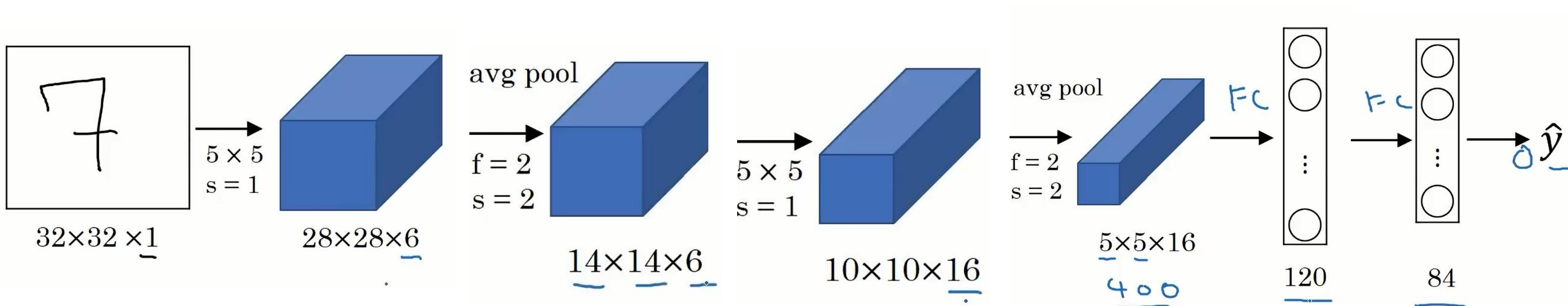
- LeNet-5 ←
- AlexNet ←
- VGG ←

ResNet

Inception



LeNet-5

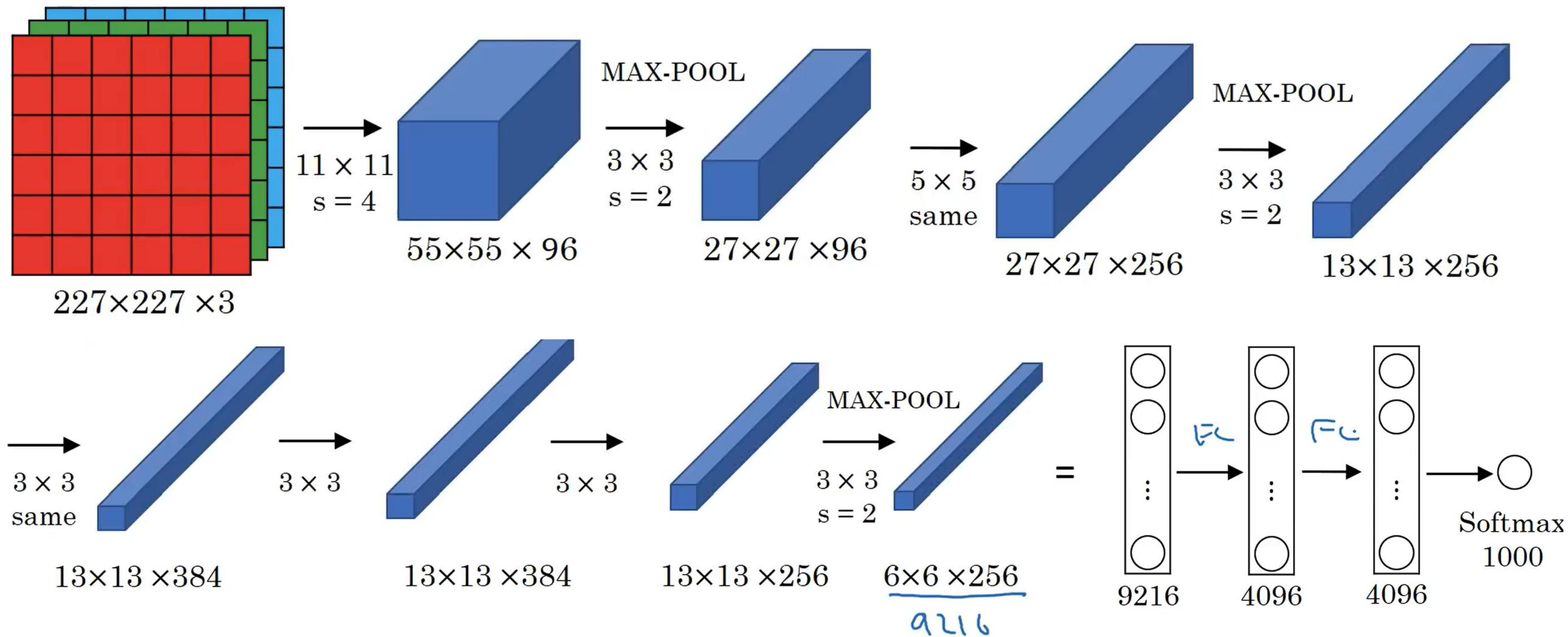


60K parameters.

$n_H, n_w \downarrow \quad n_c \uparrow$

conv pool conv pool fc fc output

AlexNet



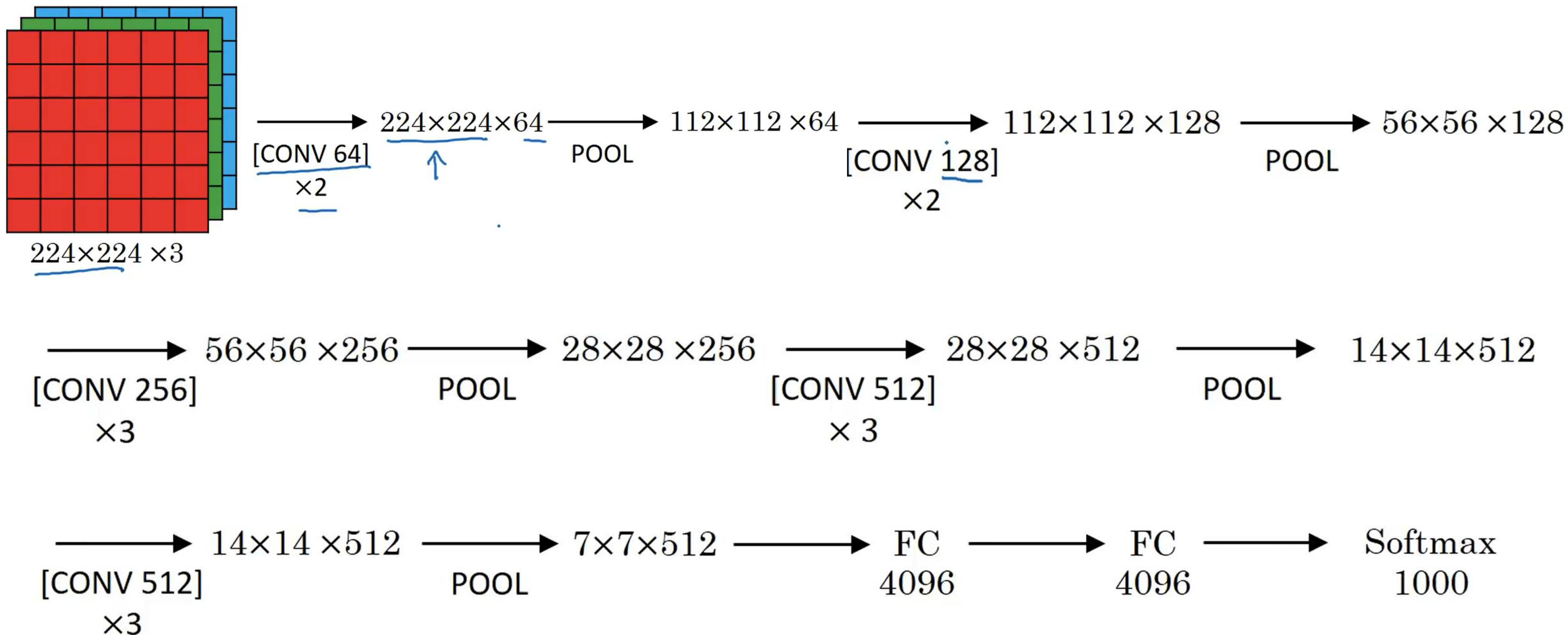
- Similar to LeNet, but much bigger.

160M parameters

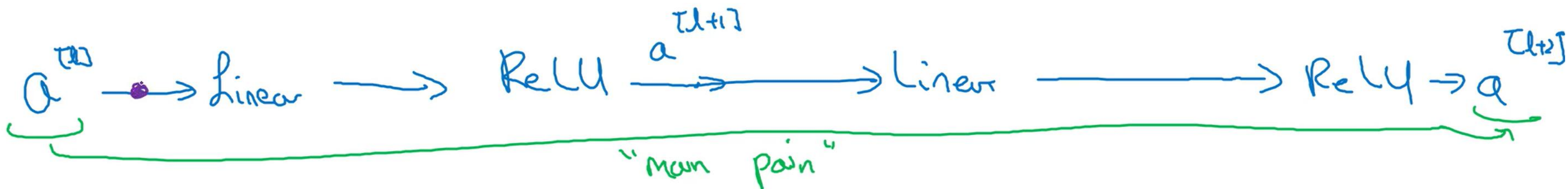
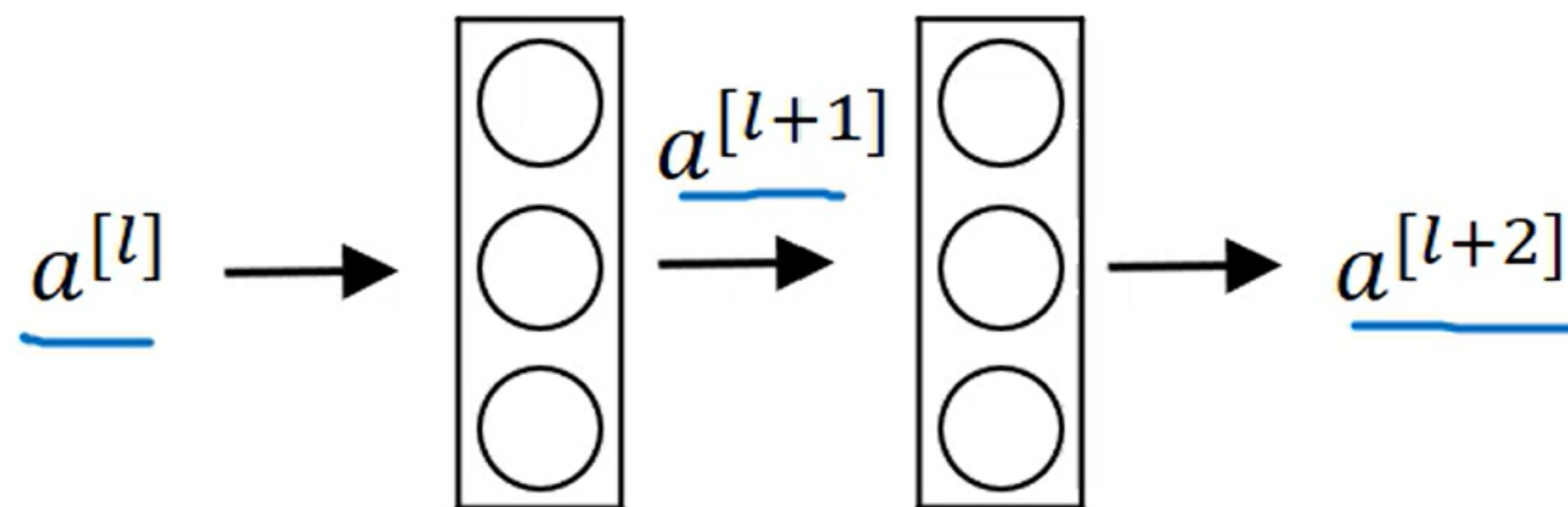
VGG-16

CONV = 3×3 filter, s = 1, same

MAX-POOL = 2×2, s = 2

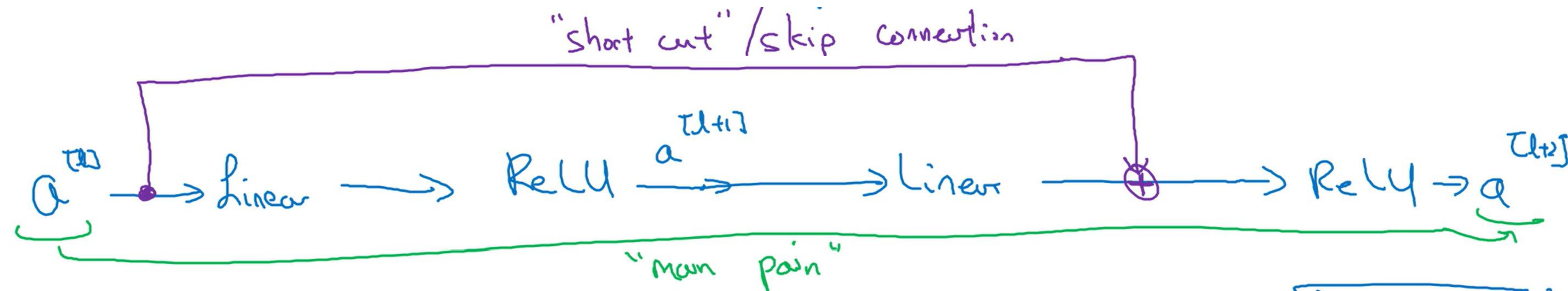


Residual Block



$$\underline{z^{[l+1]}} = W^{[l+1]} \underline{a^{[l]}} + b^{[l+1]} \quad \underline{a^{[l+1]}} = g(\underline{z^{[l+1]}}) \quad \underline{z^{[l+2]}} = W^{[l+2]} \underline{a^{[l+1]}} + b^{[l+2]} \quad \underline{a^{[l+2]}} = g(\underline{z^{[l+2]}})$$

Residual Block



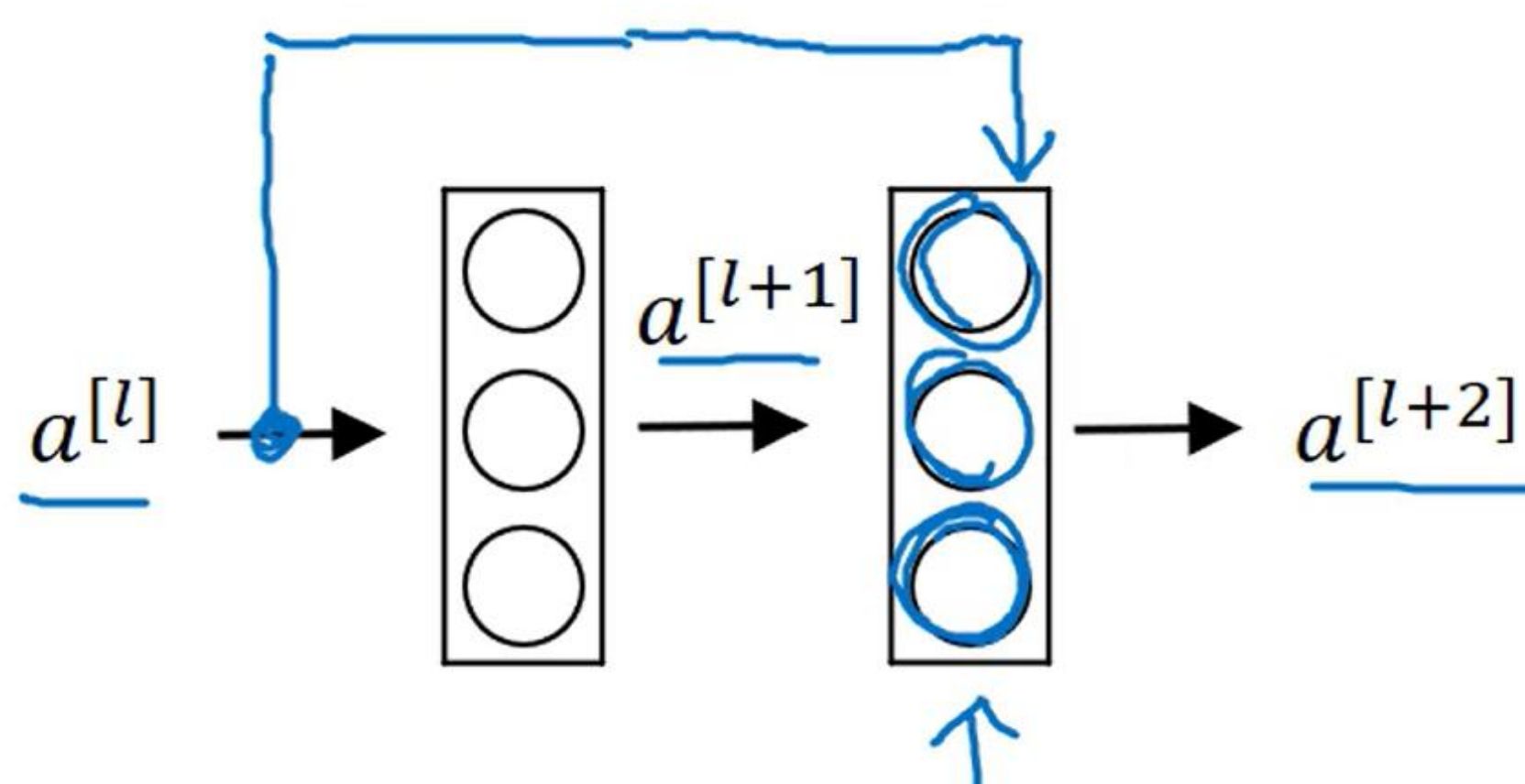
$$\underline{z^{[l+1]}} = W^{[l+1]} \underline{a^{[l]}} + b^{[l+1]}$$

$$\underline{a^{[l+1]}} = g(\underline{z^{[l+1]}})$$

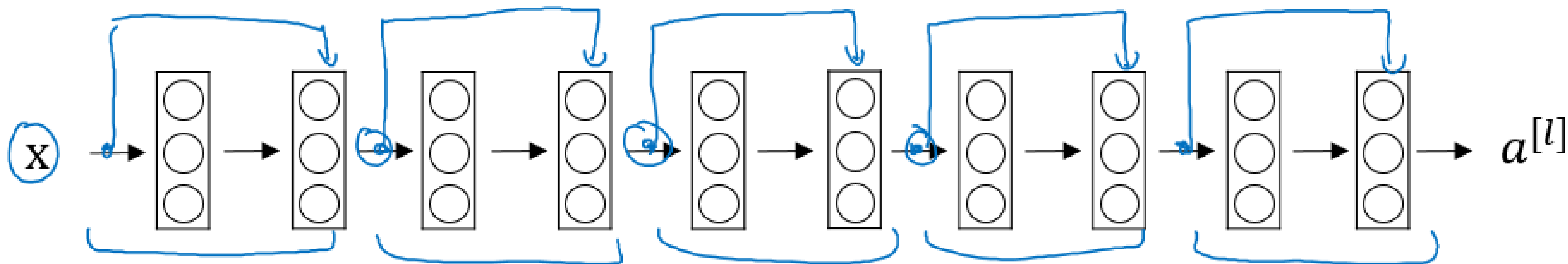
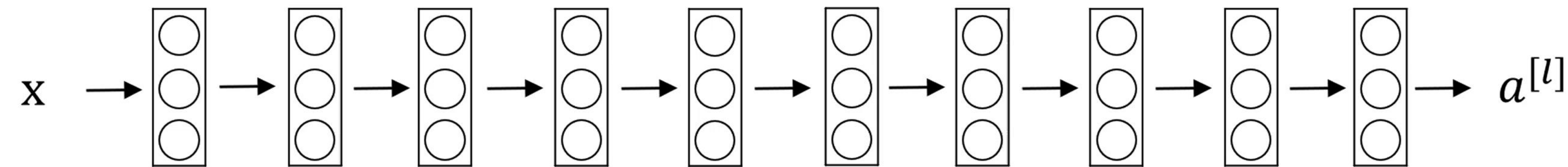
$$\underline{z^{[l+2]}} = W^{[l+2]} \underline{a^{[l+1]}} + b^{[l+2]}$$

~~$$\underline{a^{[l+2]}} = g(\underline{z^{[l+2]}})$$~~

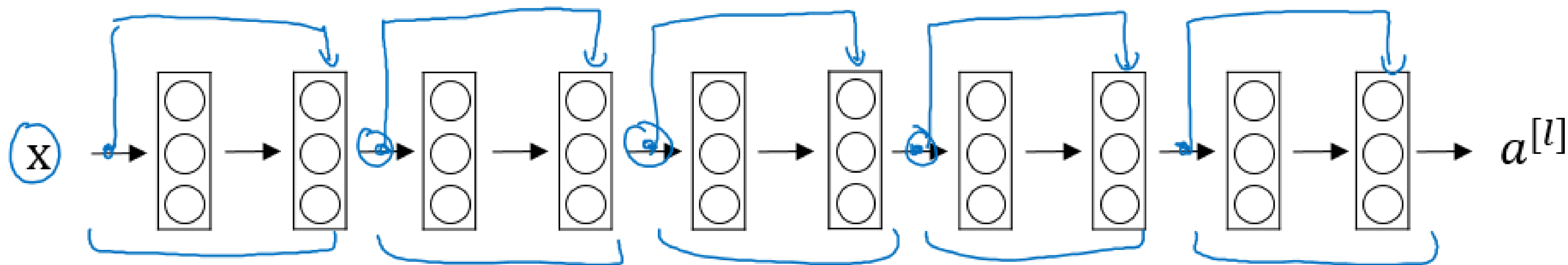
$$\underline{a^{[l+2]}} = g(\underline{z^{[l+2]}} + \underline{a^{[l]}})$$



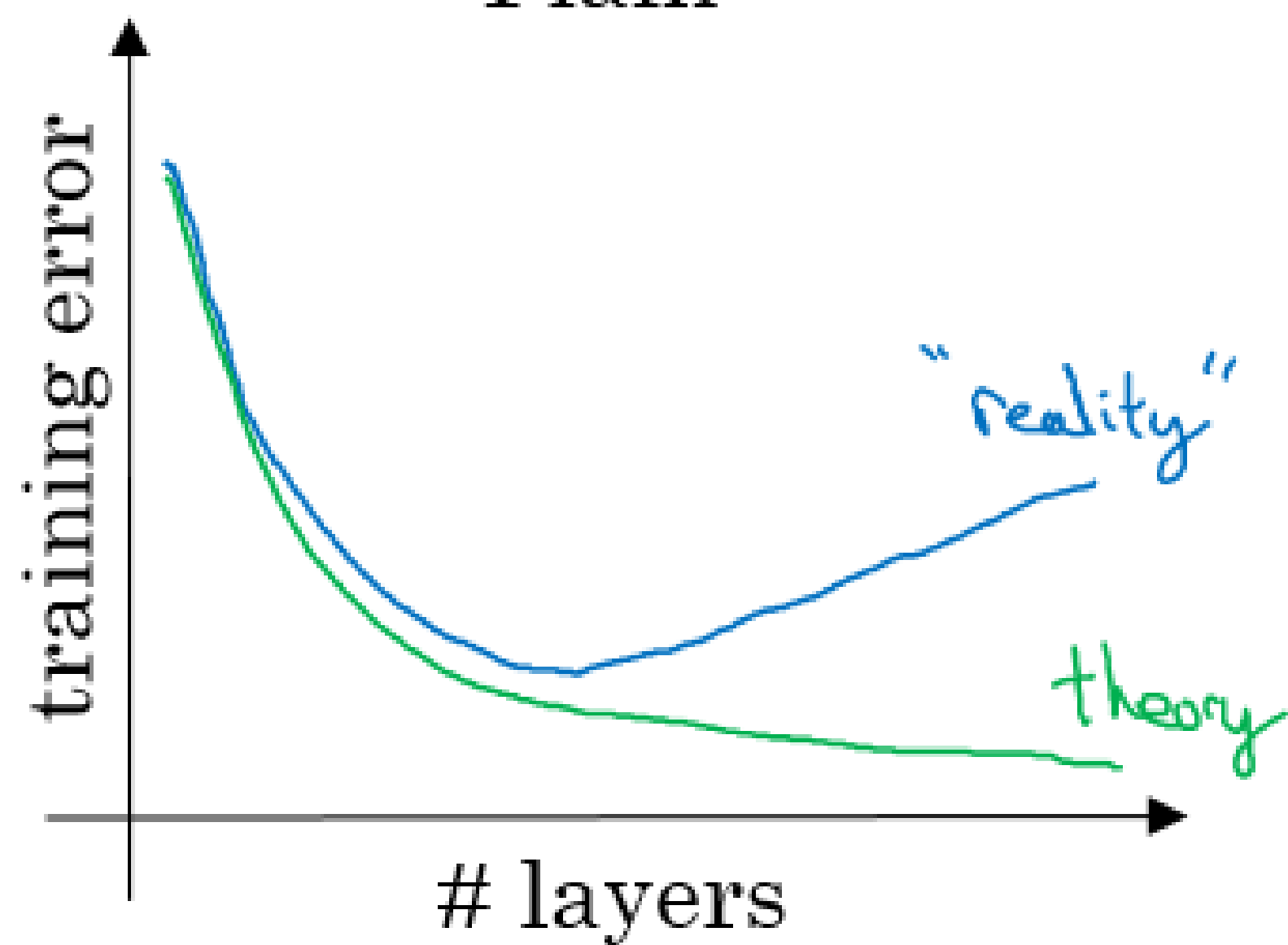
Residual Network



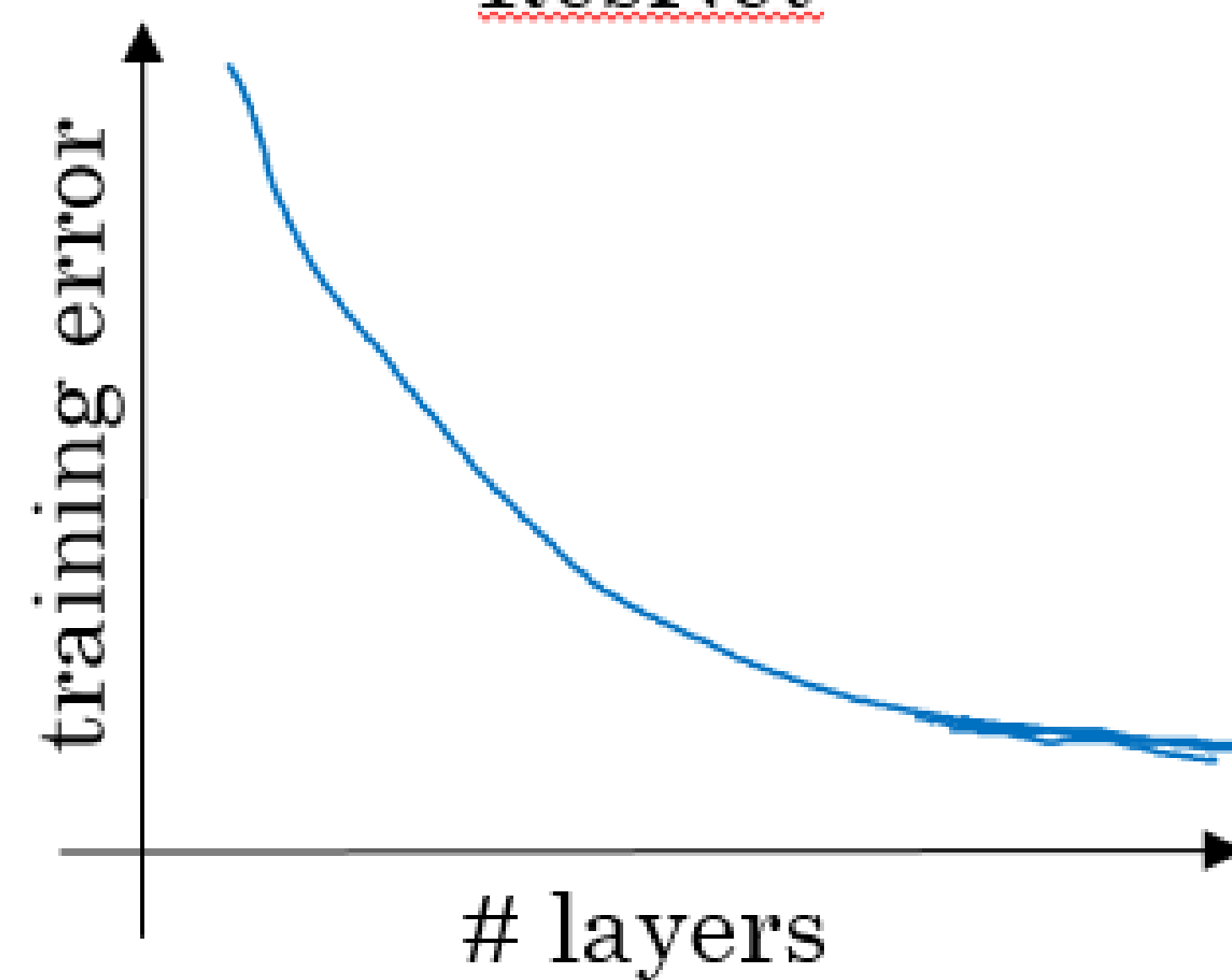
Residual Network



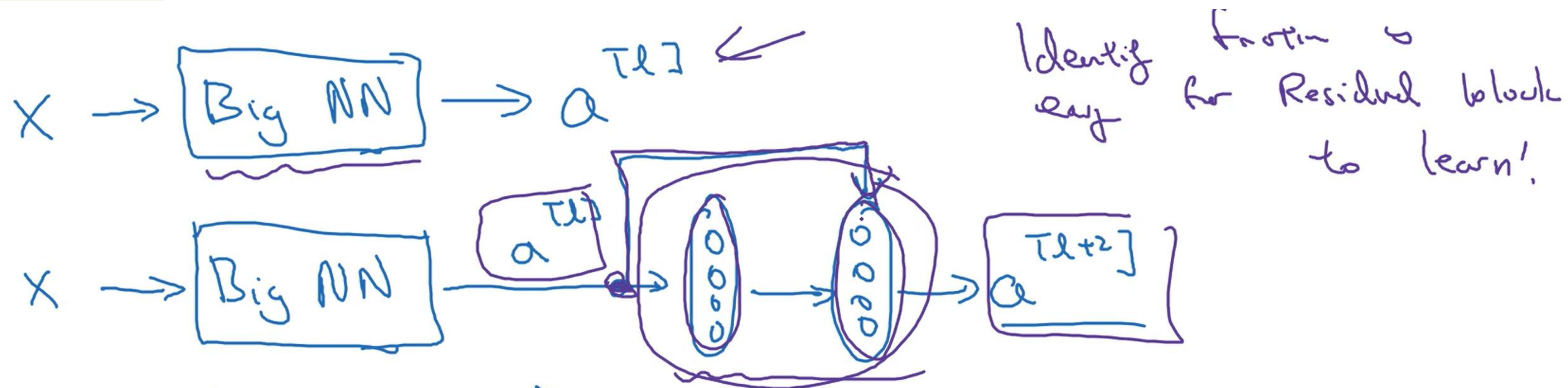
Plain



ResNet



Why do Residual Networks work



ReLU.

$$\underline{a} \geq 0$$

$$a^{[L+2]} = g\left(\underline{z}^{[L+2]} + \underline{a}^{[L]}\right)$$

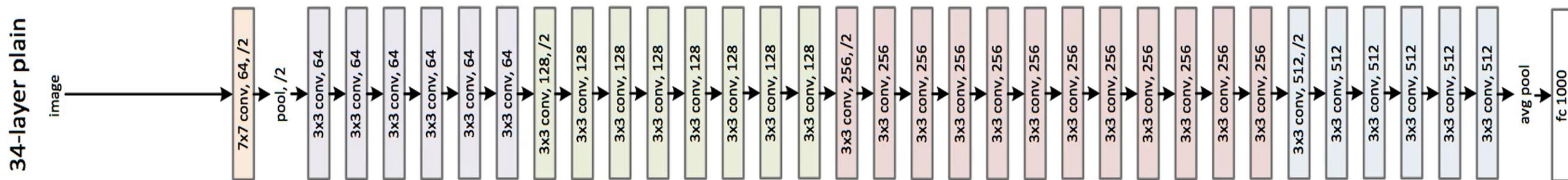
$$= g\left(\cancel{W^{[L+2]} a^{[L+1]} + b^{[L+2]}} + \underline{a}^{[L]}\right)$$

If $W^{[L+2]} = 0, b^{[L+2]} = 0$

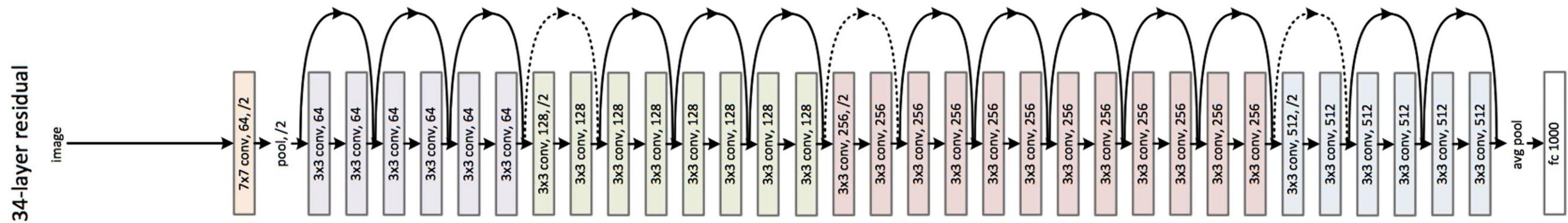
$$\begin{aligned} a^{[L]} &= g(a^{[L]}) \\ &= \underline{a}^{[L]} \end{aligned}$$

ResNet

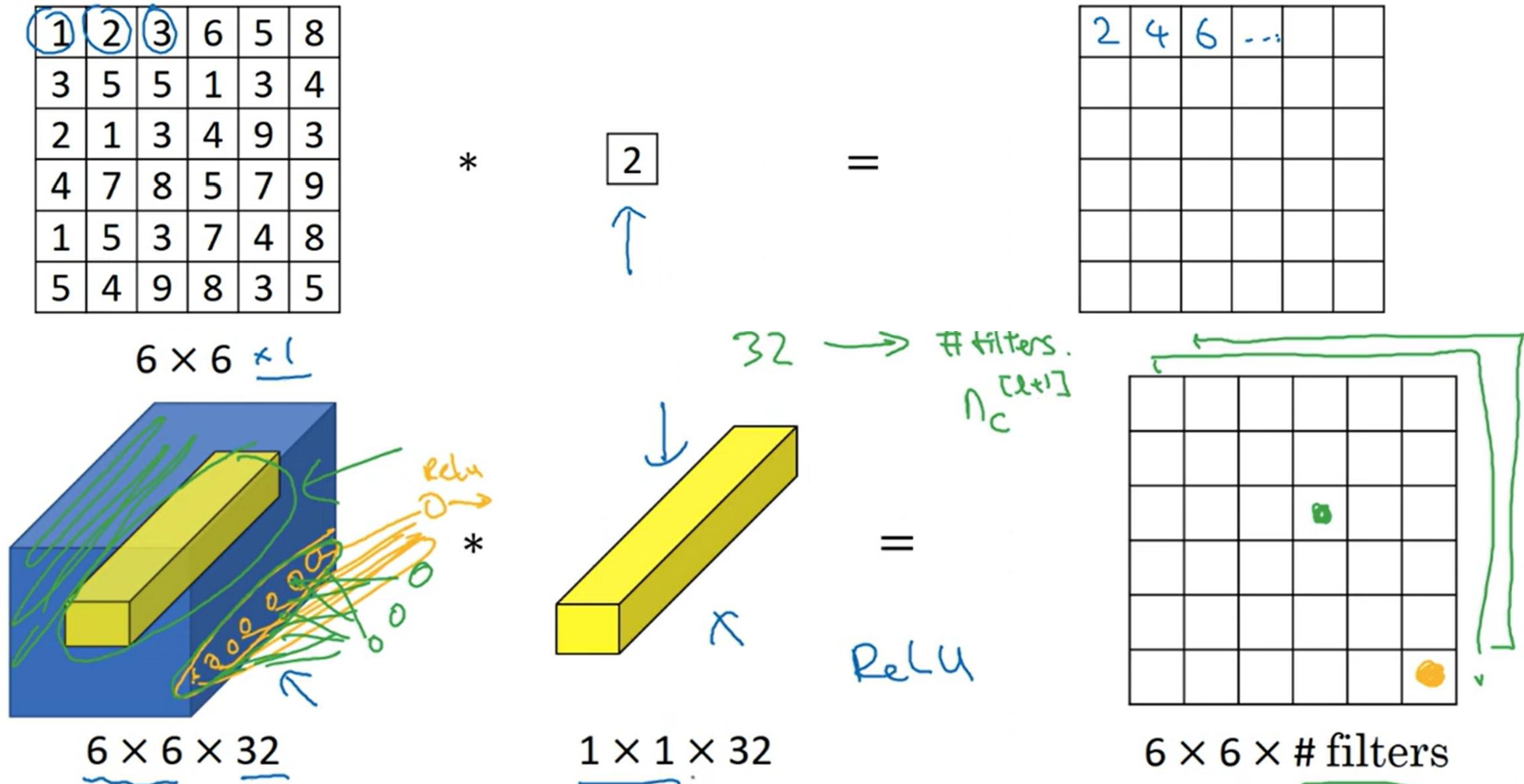
Plain



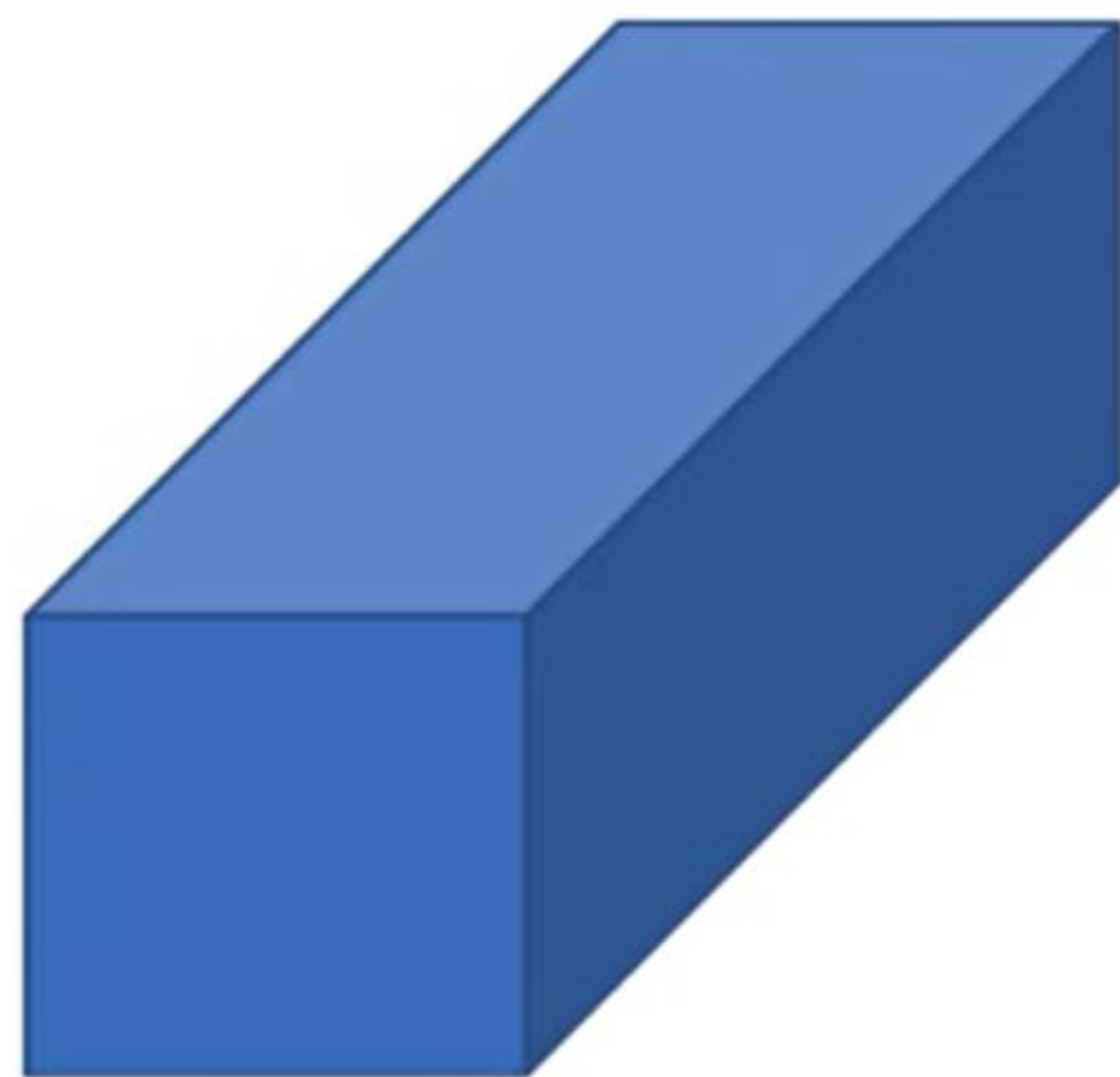
ResNet



What does a 1 X 1 convolution do



Using 1 X 1 convolution

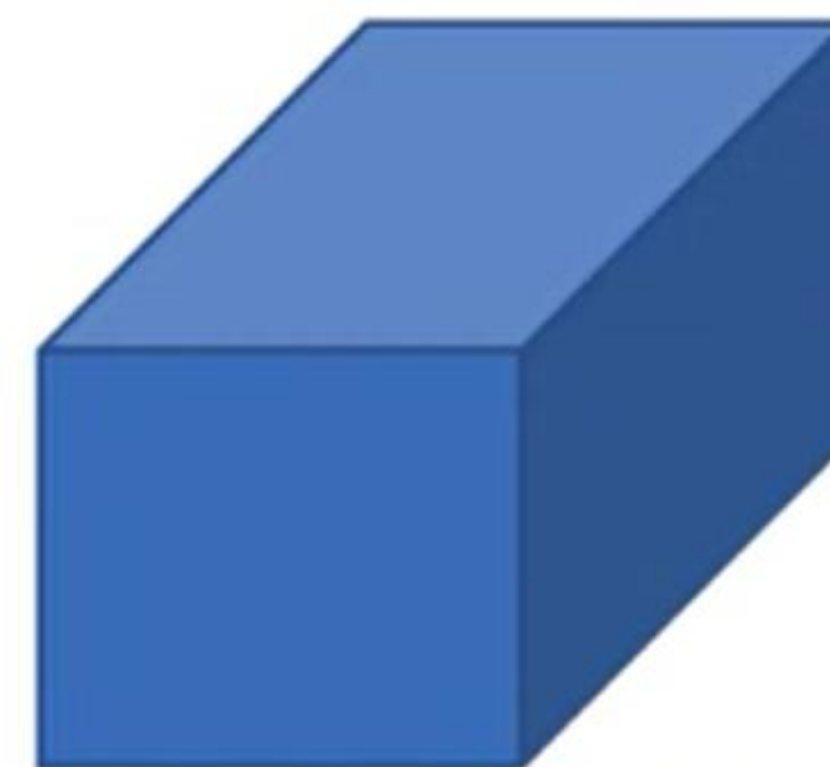


$28 \times 28 \times 192$

ReLU



CONV 1×1
32



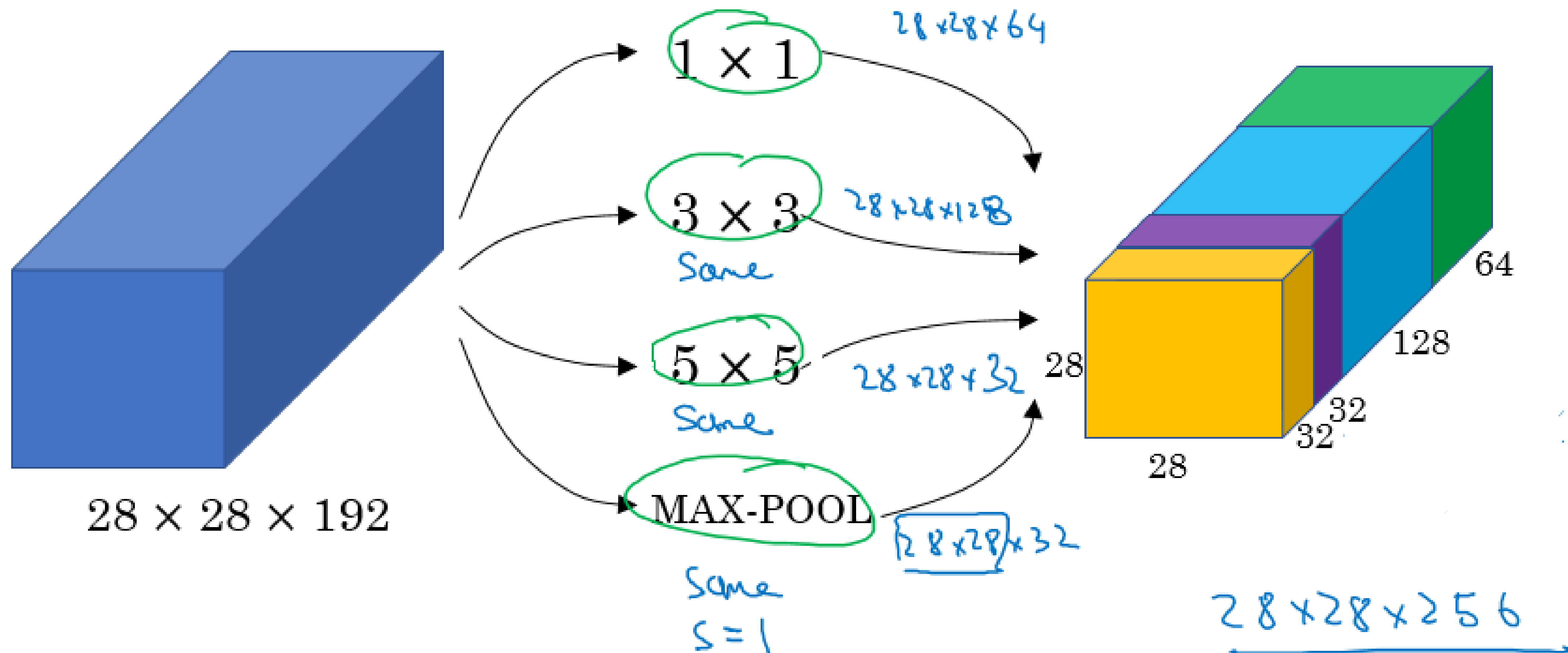
$28 \times 28 \times 32$

$1 \times 1 \times 192$

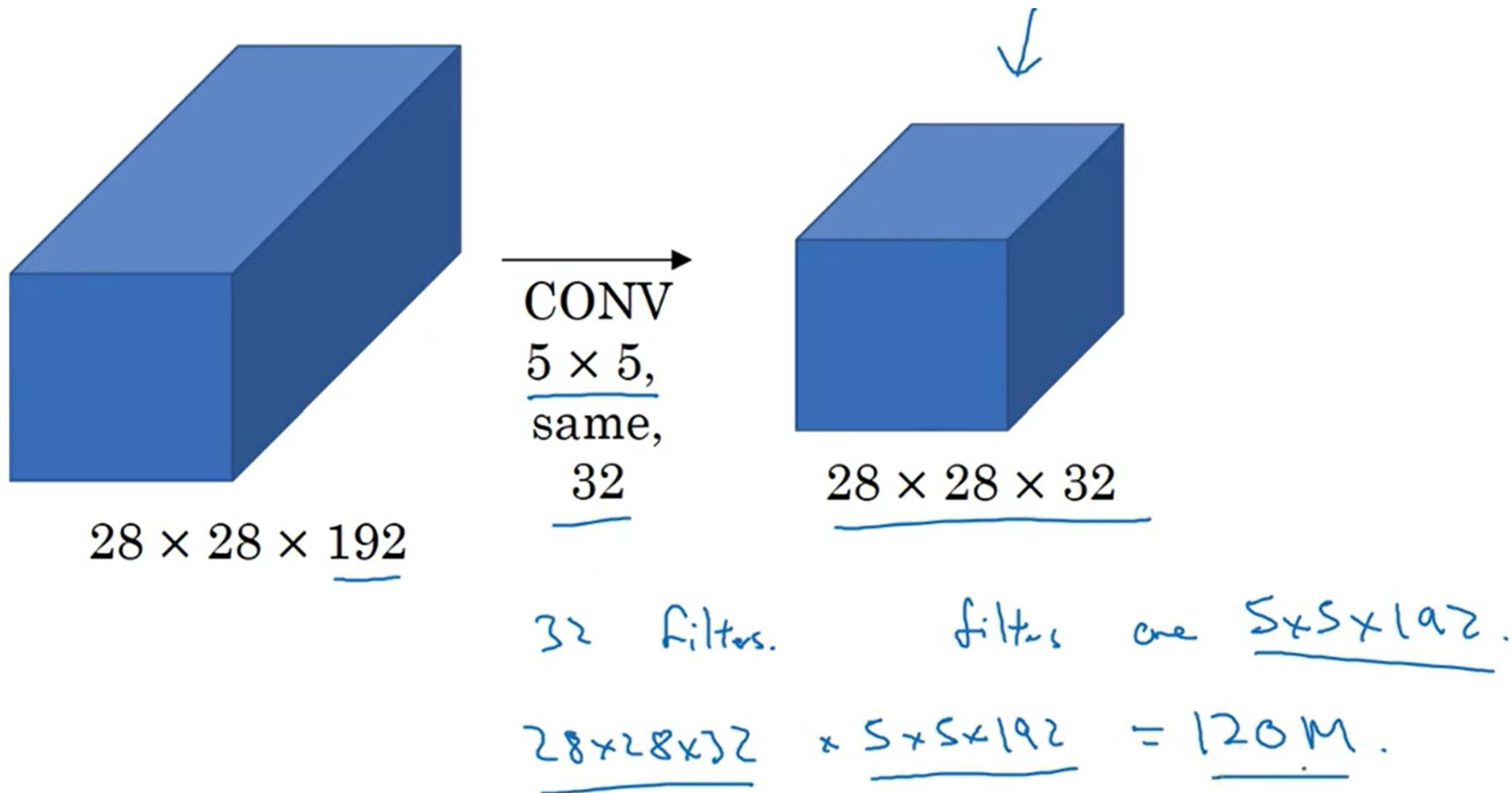
32 filters.

n_H, n_W, n_C

Inception network motivation

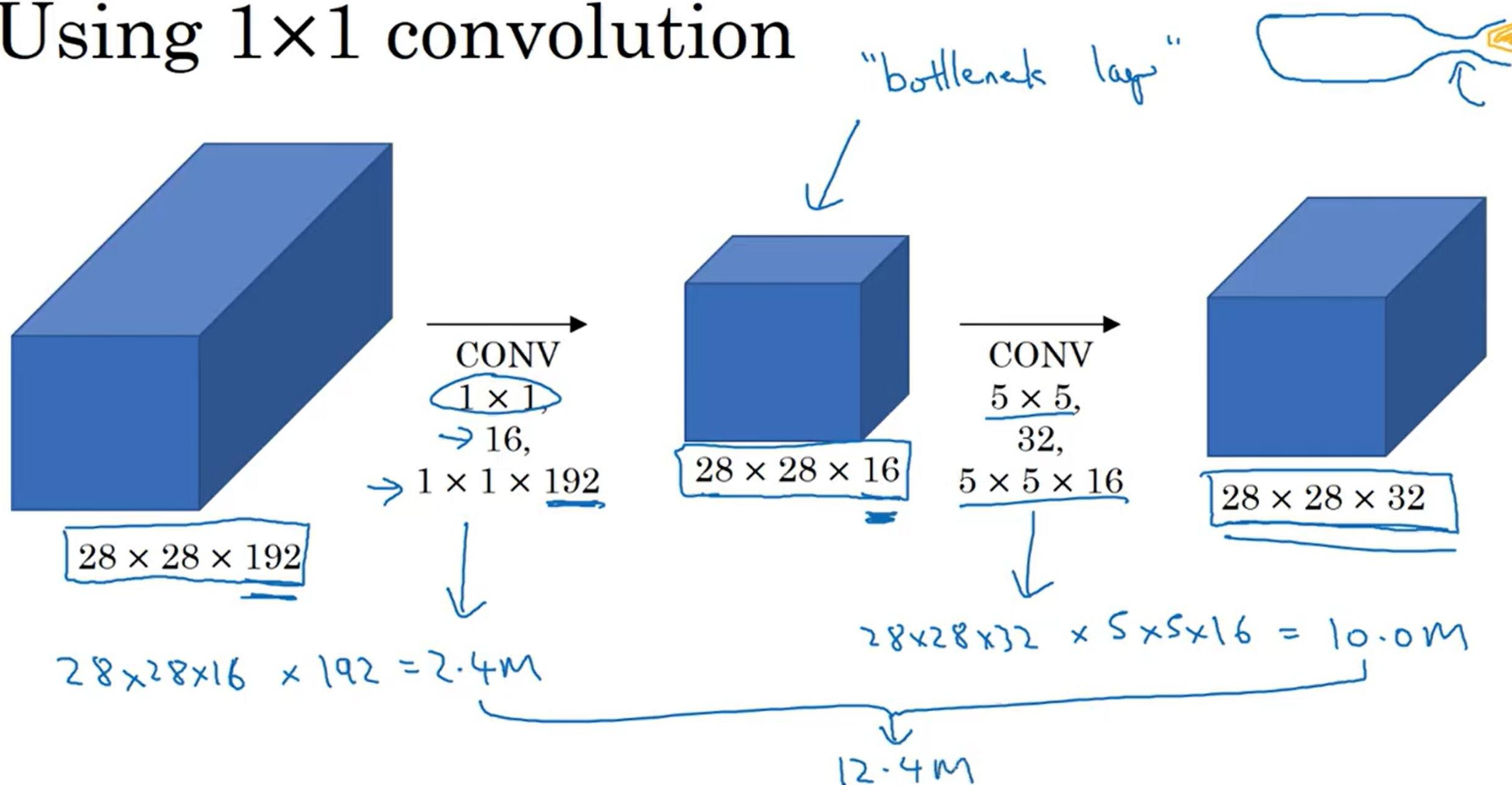


Problem of computational cost

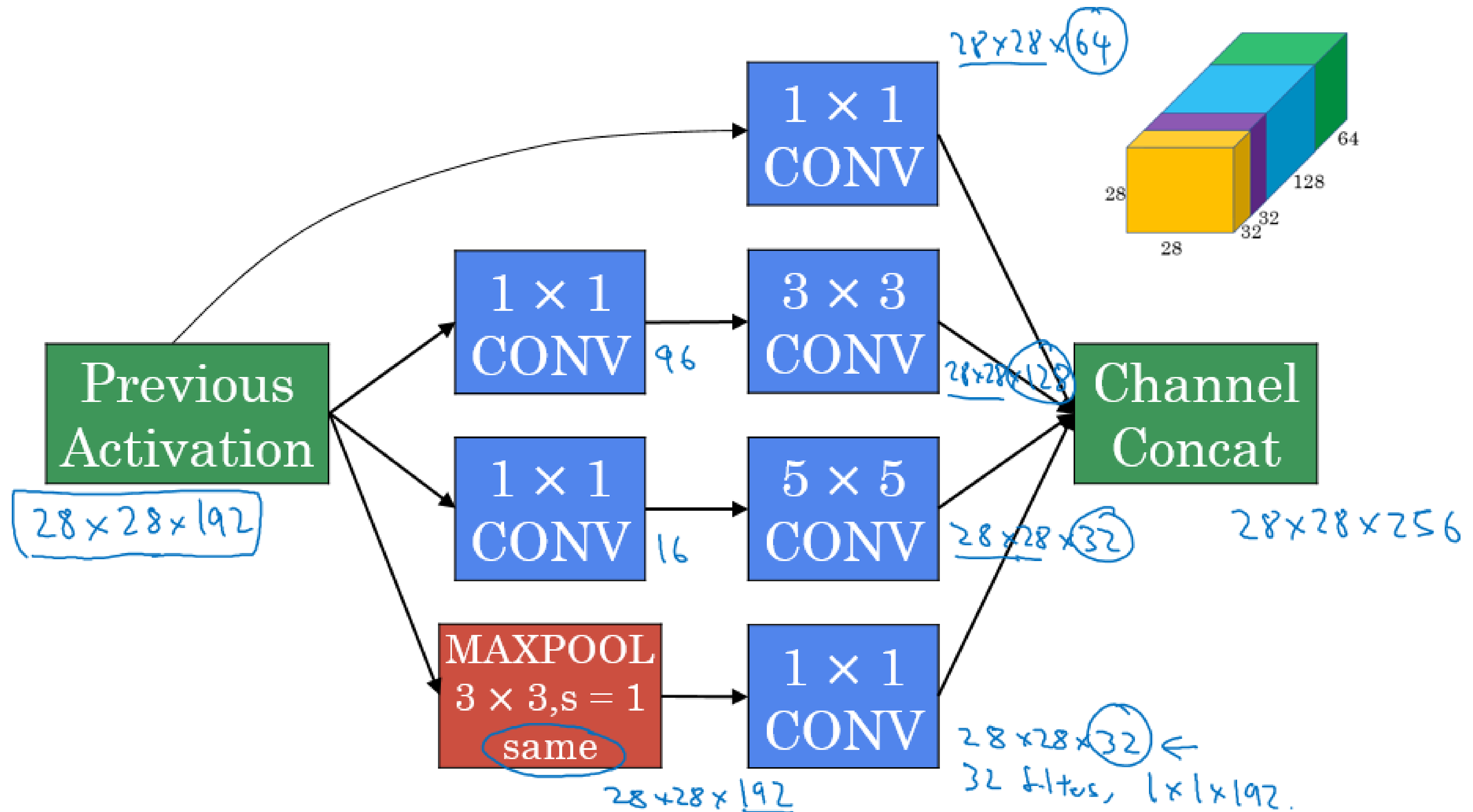


Problem of computational cost

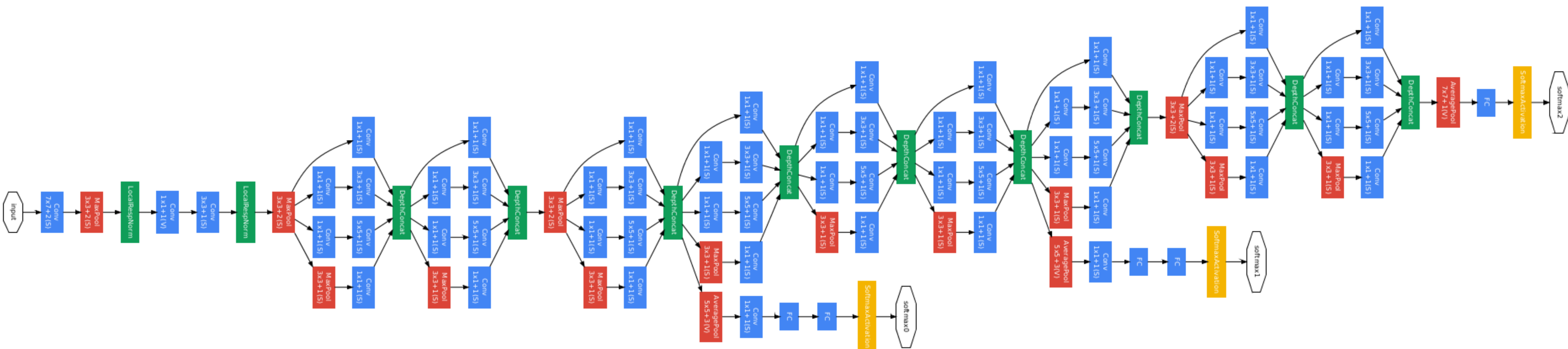
Using 1×1 convolution



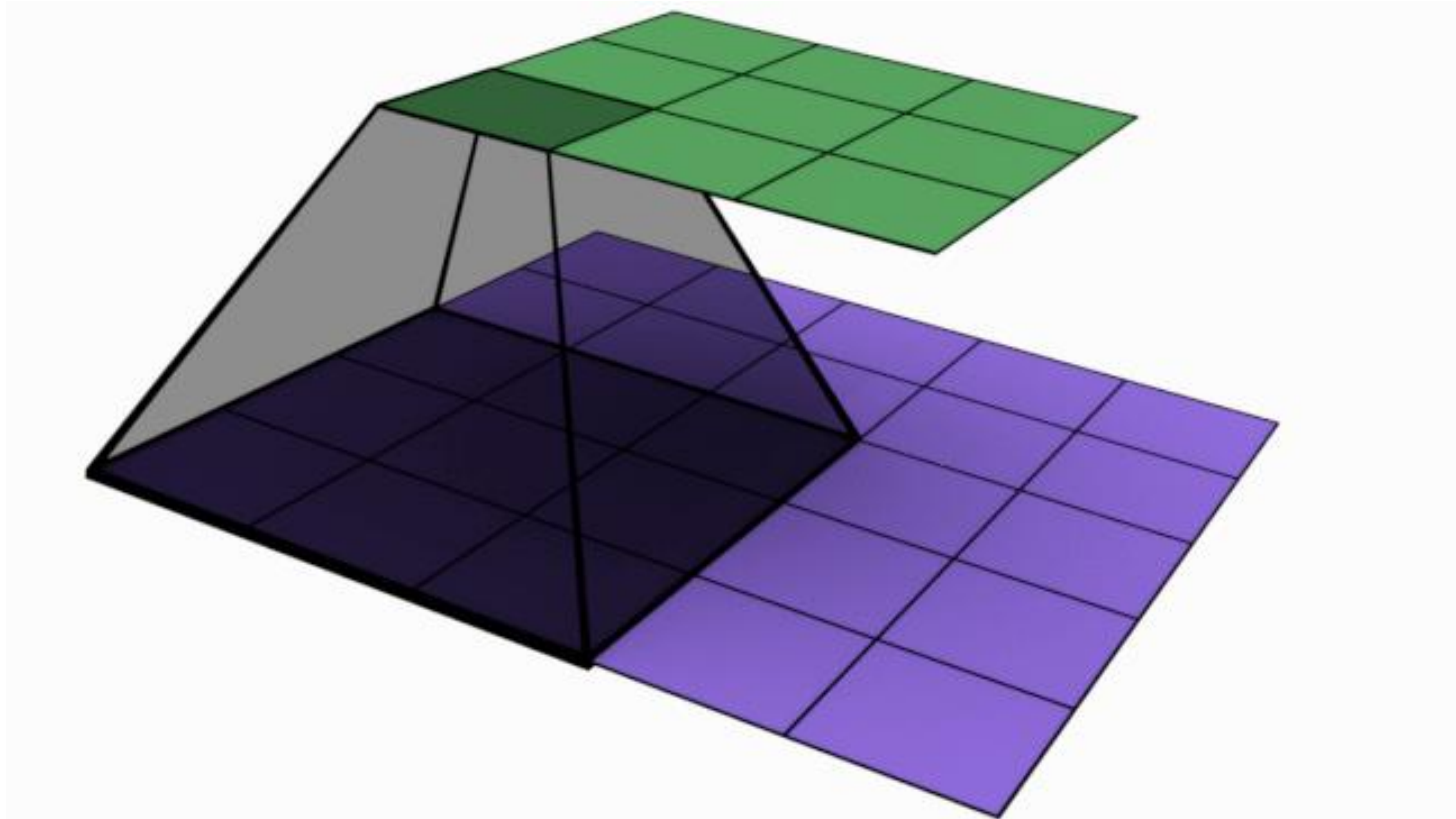
Inception module



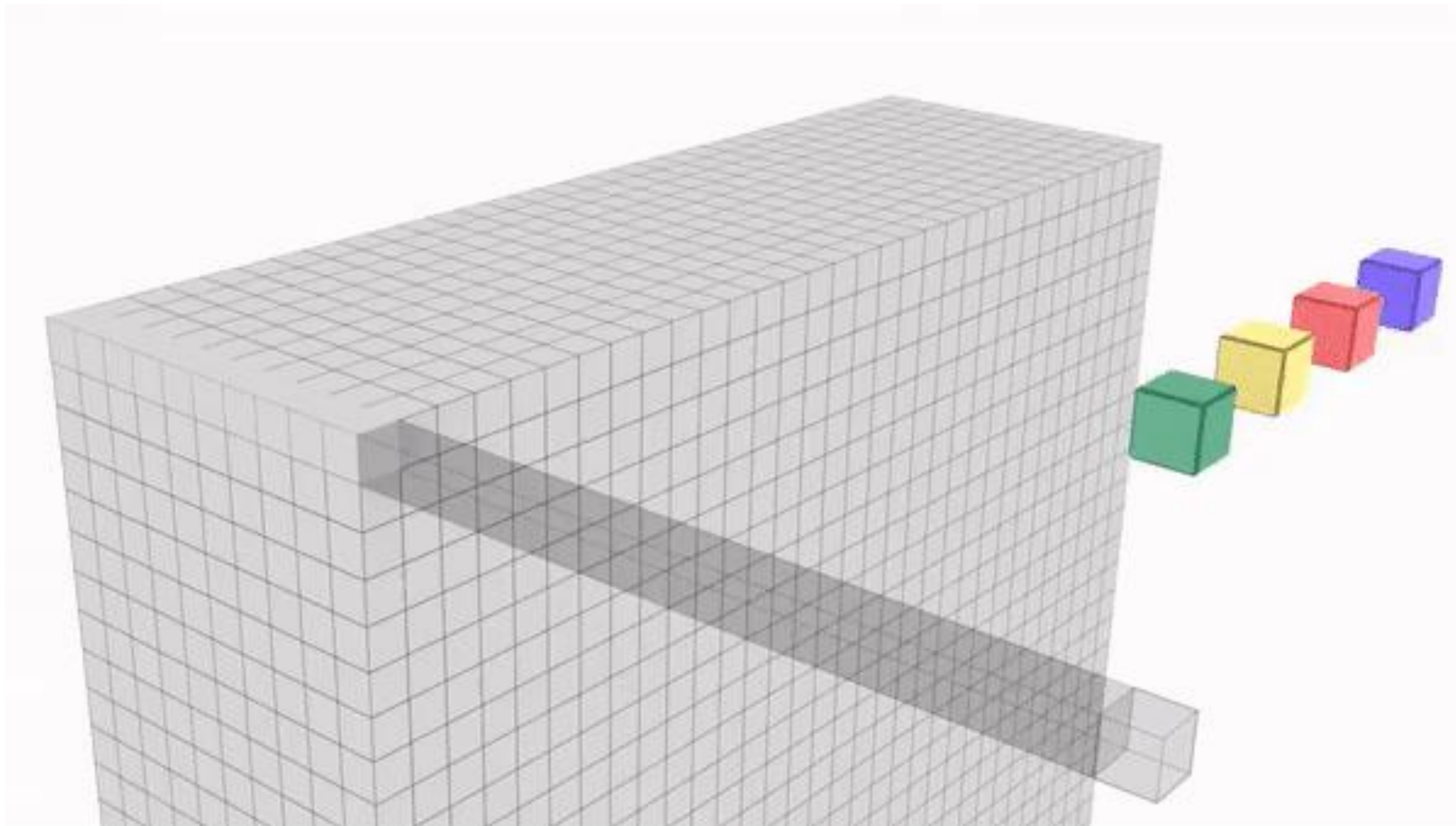
Inception network



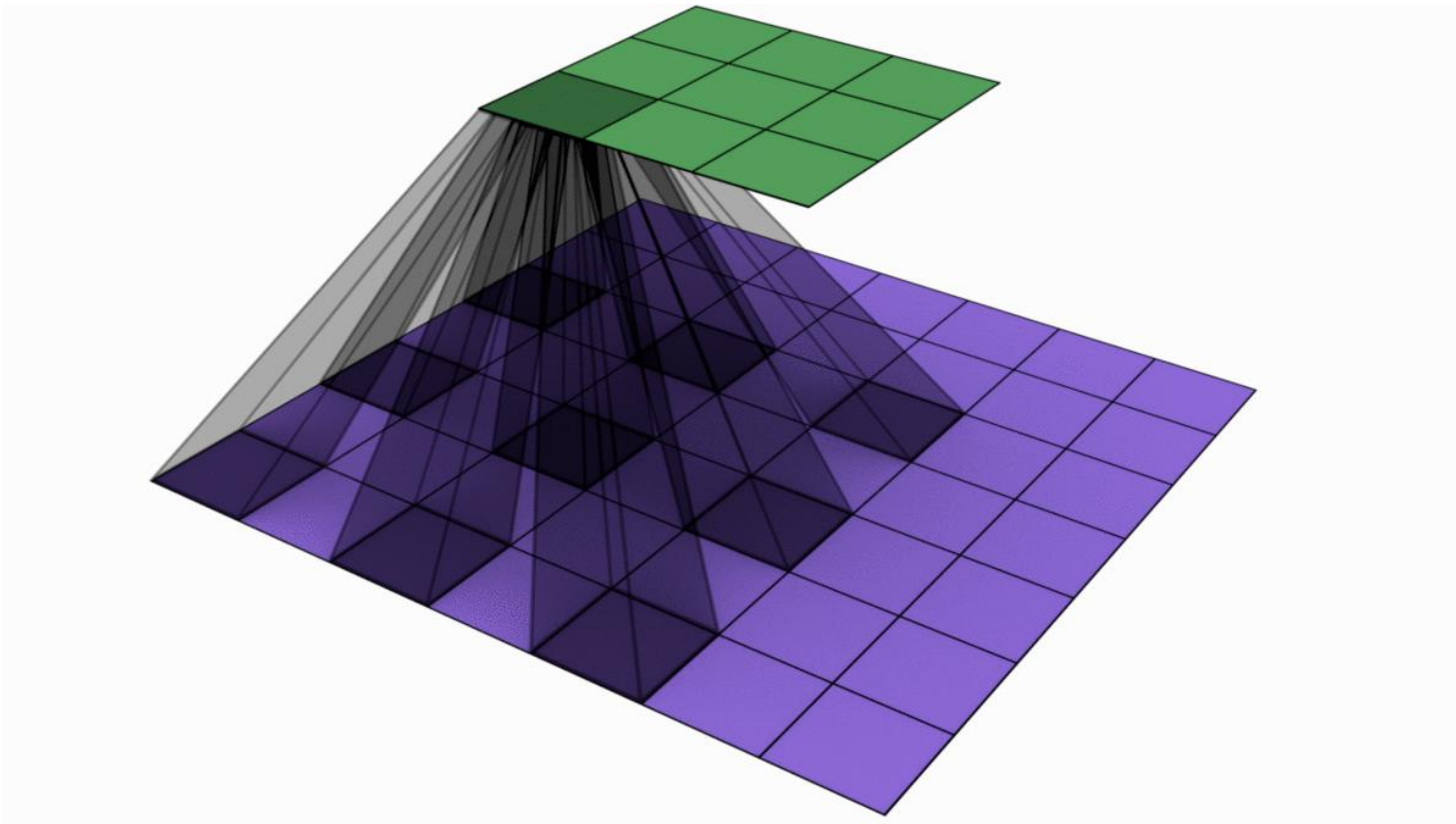
Normal Convolutions | 3x3 Convolution



Pointwise Convolutions | 1x1 Convolutions

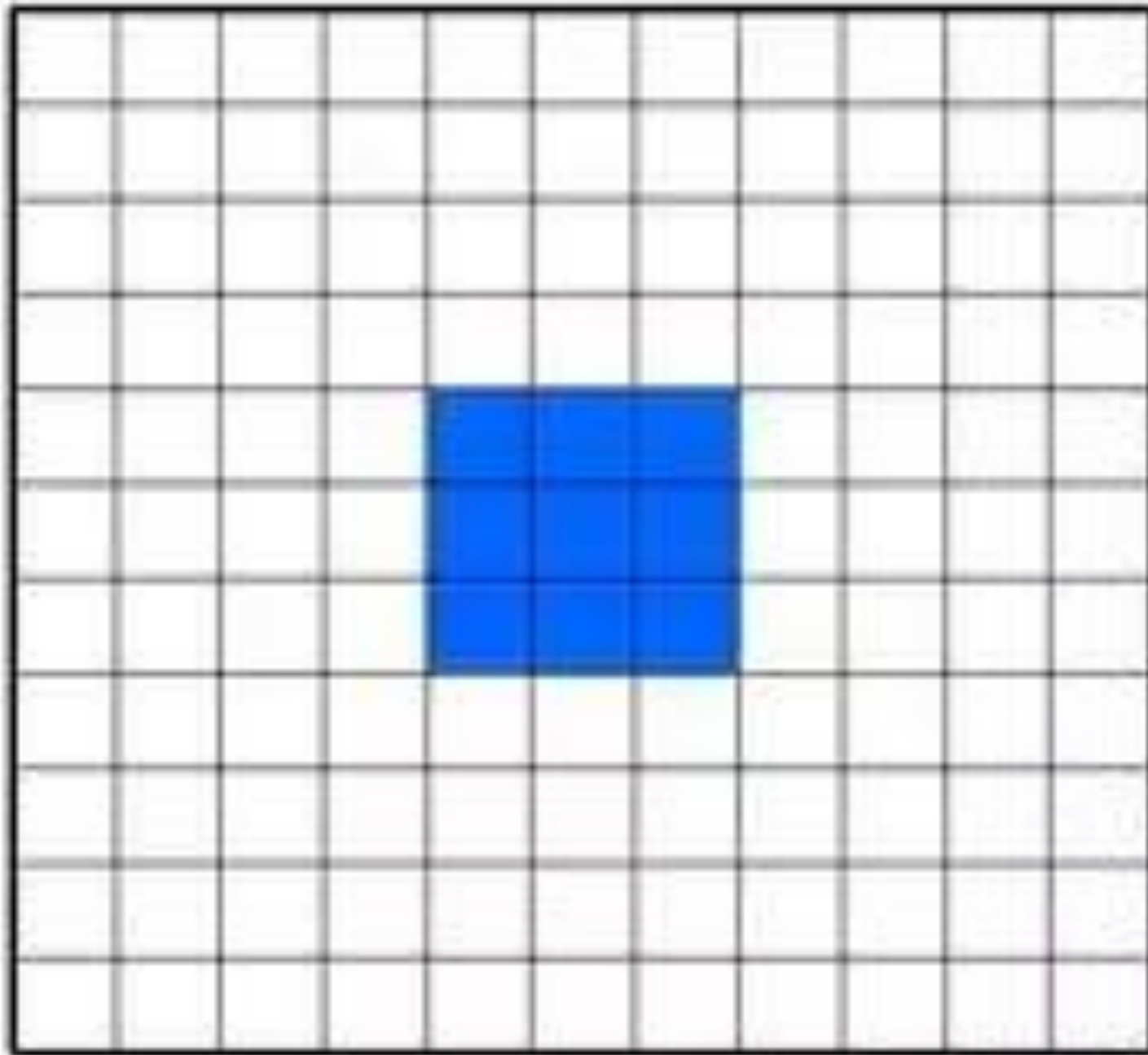


Dilated convolutions | Atrous Convolutions

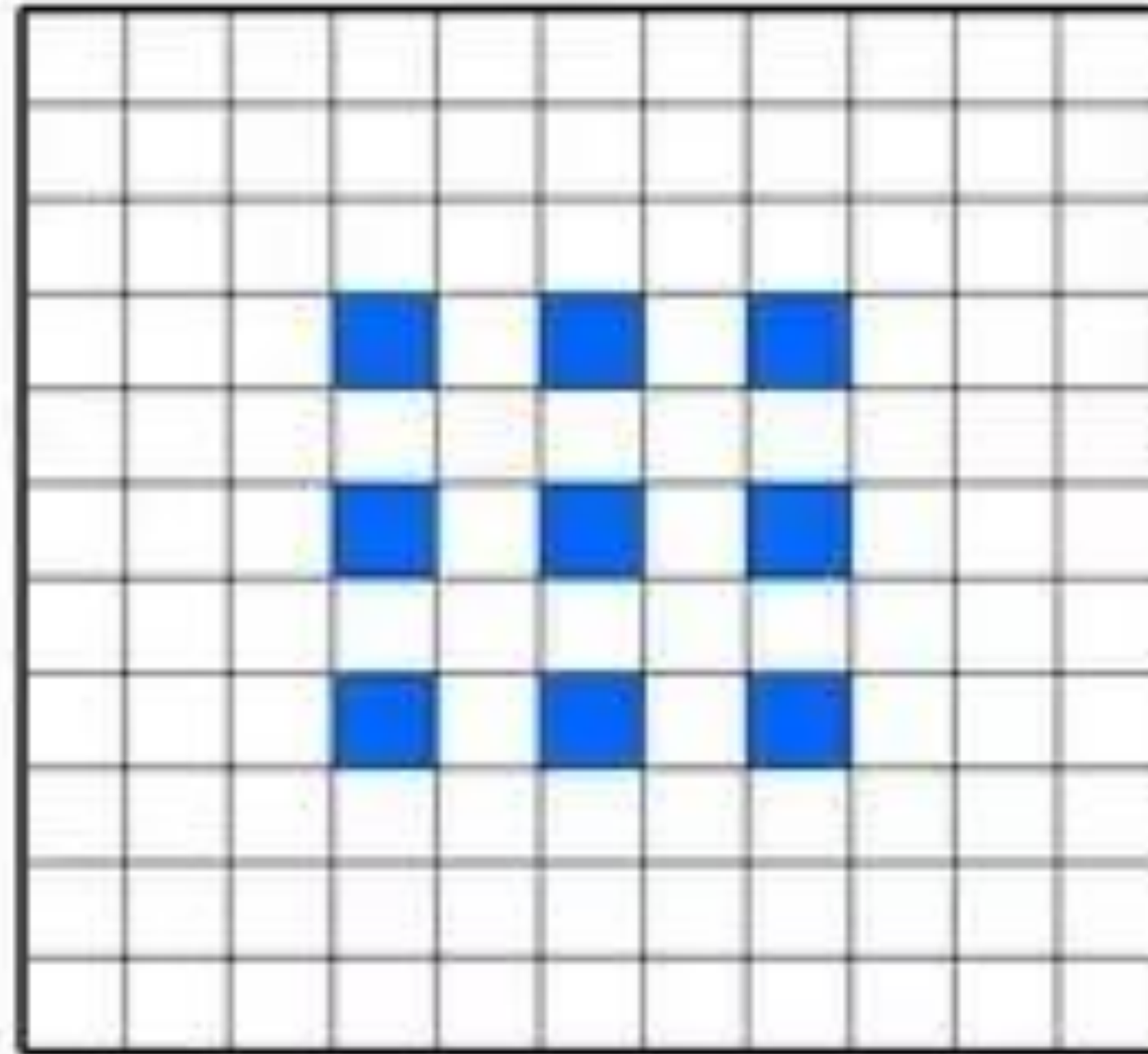


Dilated convolutions | Atrous Convolutions

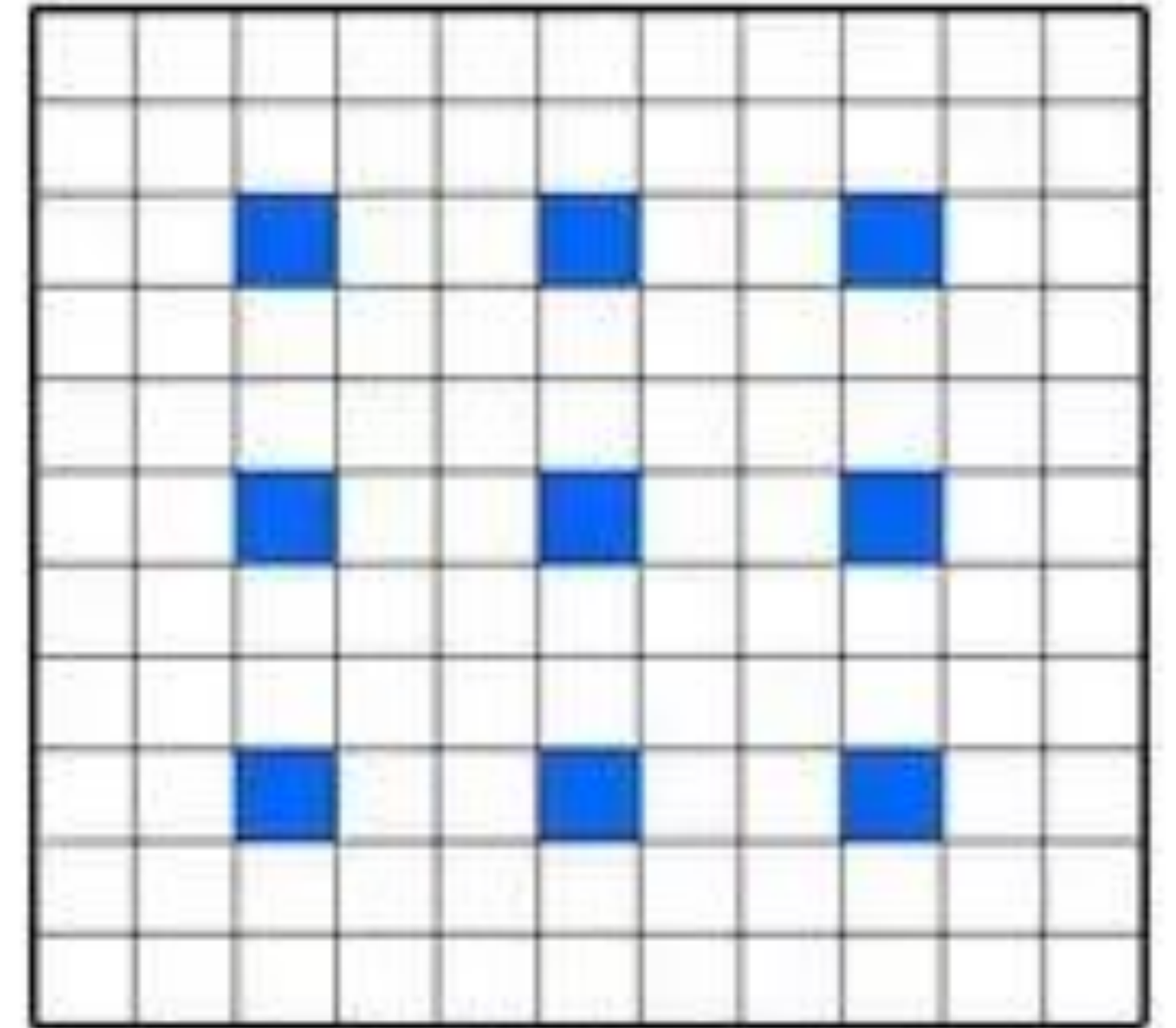
$D=1$



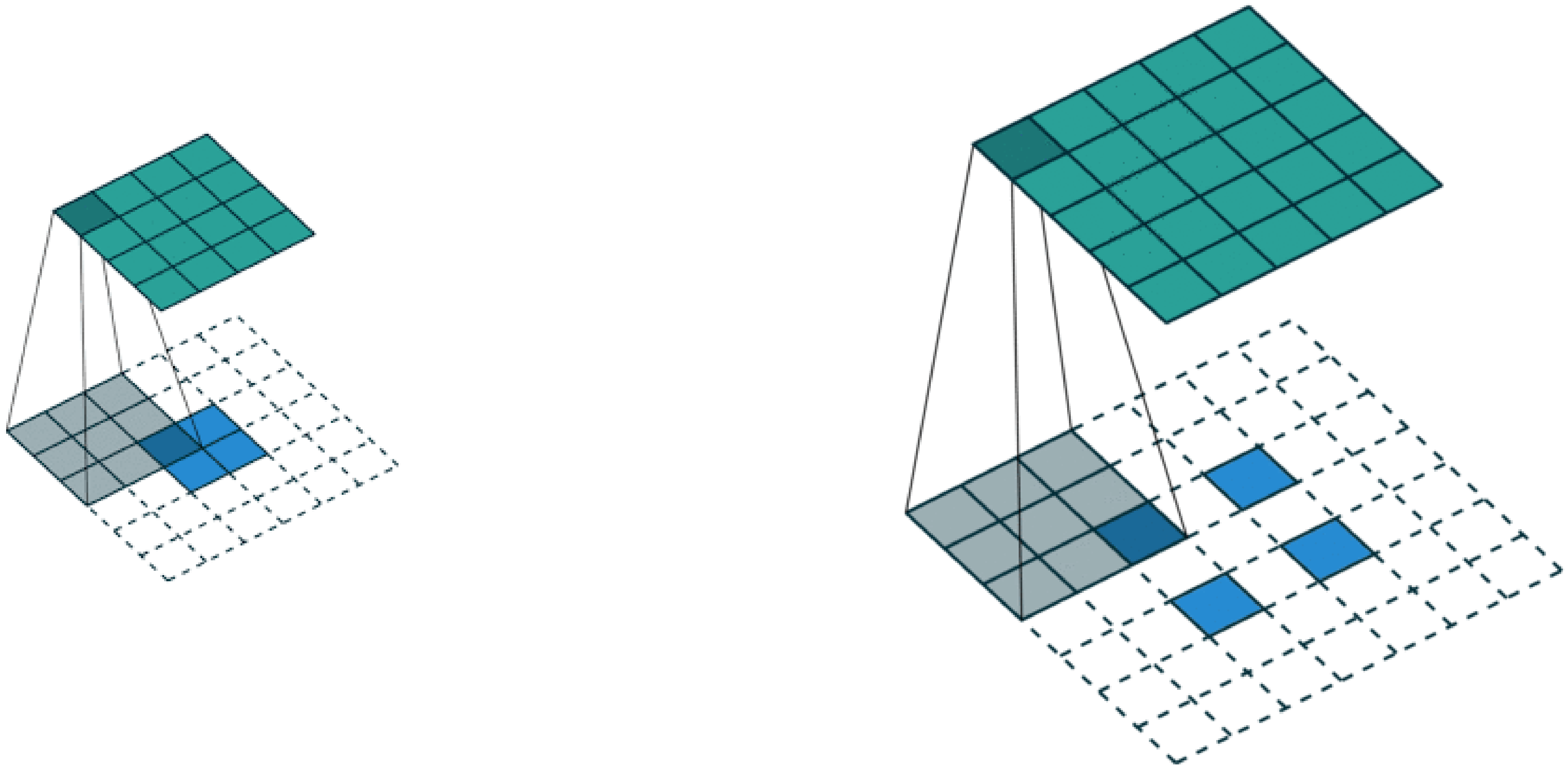
$D=2$



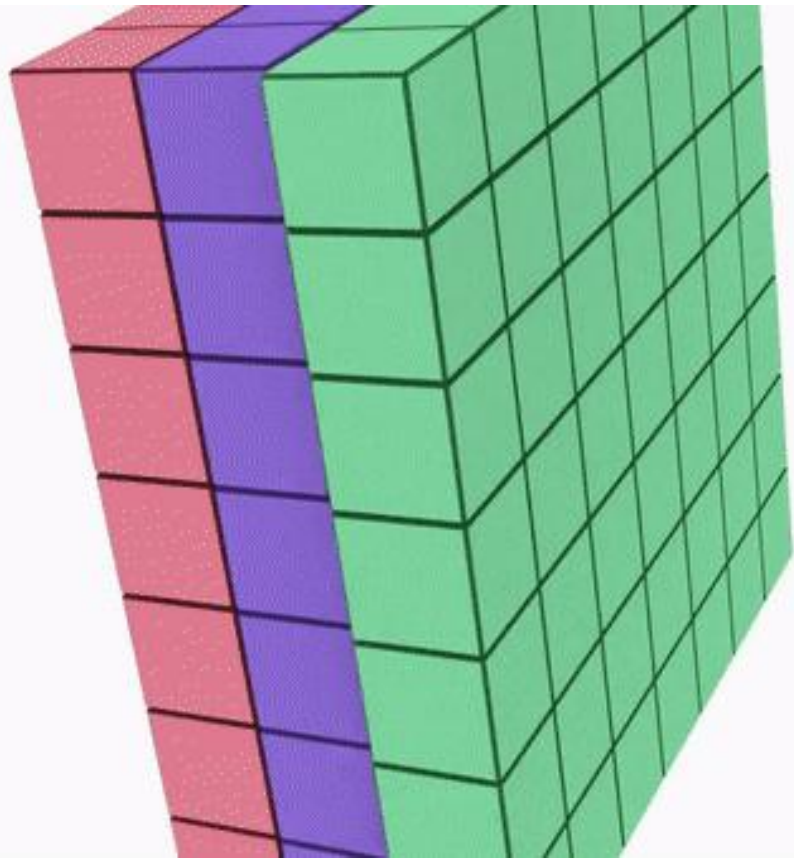
$D=3$



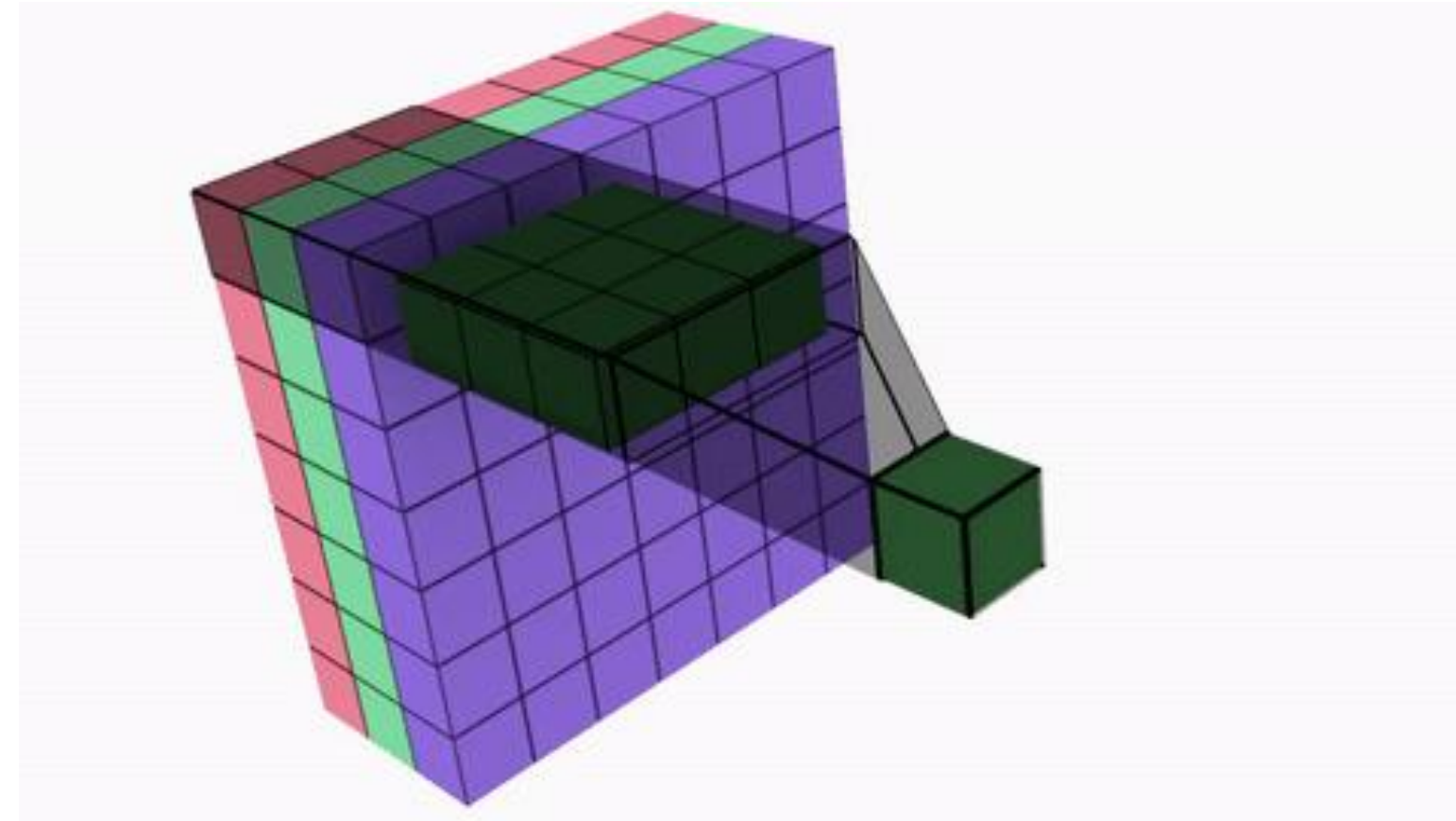
Deconvolution | Transpose convolution



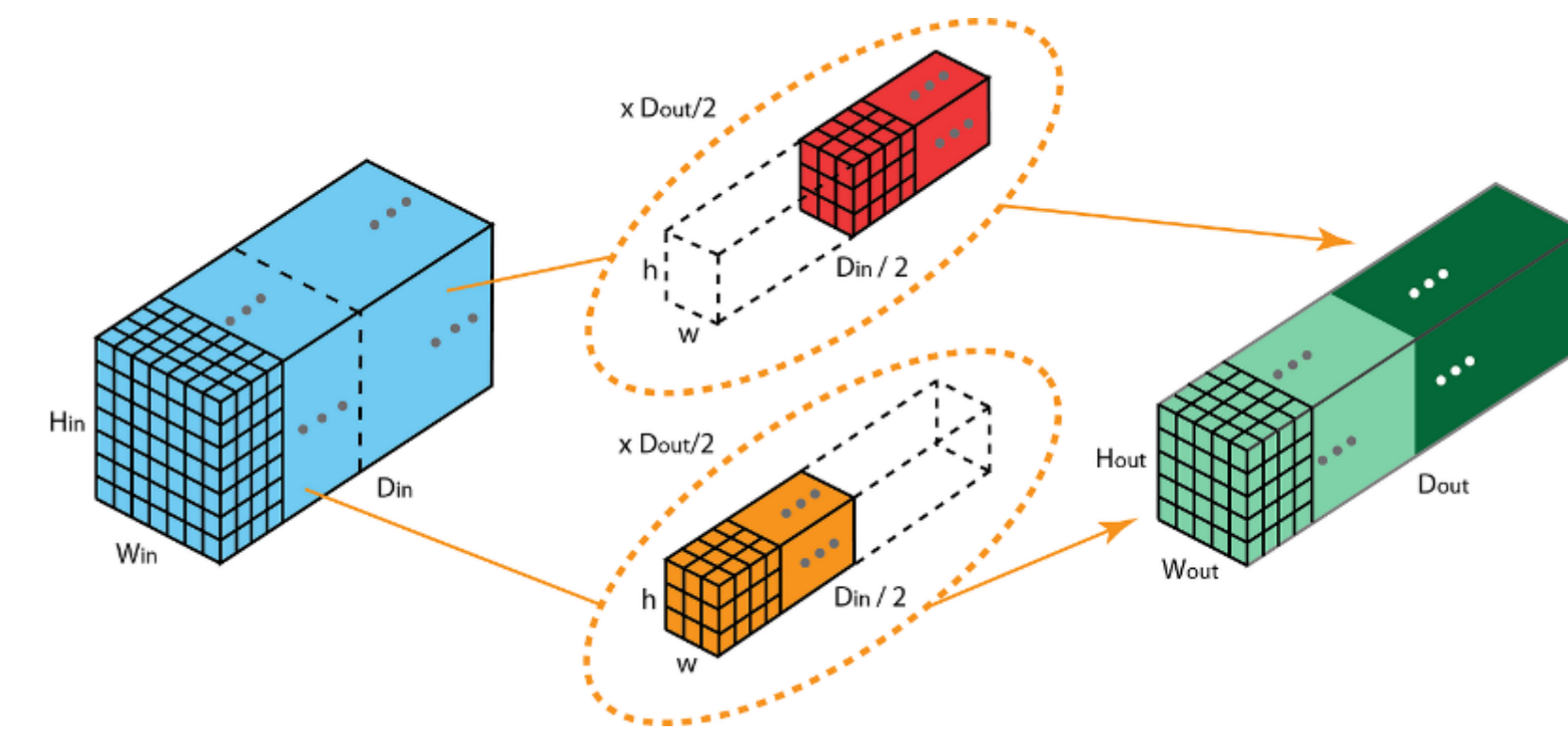
More...



Depthwise convolution animation



Spatially Separable Convolution



Grouped convolution

Thank You

Any questions?

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Source & Special Thanks to Andrew Ng (Coursera)

<https://www.youtube.com/@Deeplearningai>

<https://www.youtube.com/playlist?list=PLkDaE6sCZn6Gl29AoE31iwdVwSG-KnDzF>

<https://medium.com/hitchhikers-guide-to-deep-learning/10-introduction-to-deep-learning-with-computer-vision-types-of-convolutions-atrous-convolutions-3cf142f77bc0>